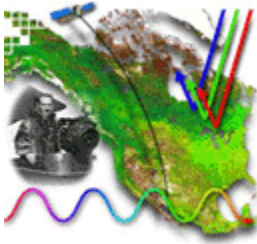


Fundamentals of Remote Sensing



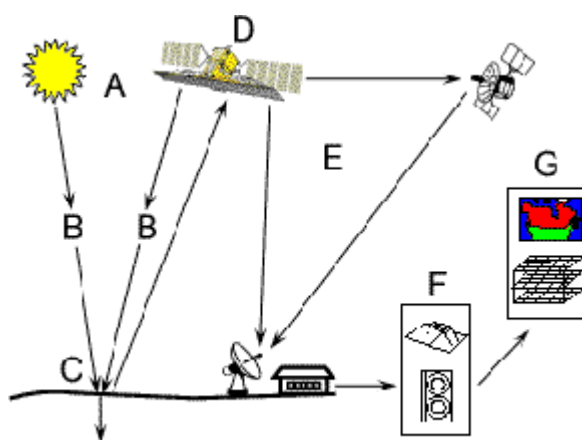
1. Introduction to Fundamentals

1.1 What is Remote Sensing?

So, what exactly is **remote sensing**? For the purposes of this tutorial, we will use the following definition:

"Remote sensing is the science (and to some extent, art) of acquiring information about the Earth's surface without actually being in contact with it. This is done by sensing and recording reflected or emitted energy and processing, analyzing, and applying that information."

In much of remote sensing, **the process** involves an interaction between incident radiation and the targets of interest. This is exemplified by the use of imaging systems where the following seven elements are involved. Note, however that remote sensing also involves the sensing of emitted energy and the use of non-imaging sensors.



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1. Energy Source or Illumination (A) - the first requirement for remote sensing is to have an energy source which illuminates or provides electromagnetic energy to the target of interest.

2. Radiation and the Atmosphere (B) - as the energy travels from its source to the target, it will come in contact with and interact with the atmosphere it passes through. This interaction may take place a second time as the energy travels from the target to the sensor.

3. Interaction with the Target (C) - once the energy makes its way to the target through the atmosphere, it interacts with the target depending on the properties of both the target and the radiation.

Section 1.1 What is Remote Sensing?

4. **Recording of Energy by the Sensor (D)** - after the energy has been scattered by, or emitted from the target, we require a sensor (remote - not in contact with the target) to collect and record the electromagnetic radiation.

5. **Transmission, Reception, and Processing (E)** - the energy recorded by the sensor has to be transmitted, often in electronic form, to a receiving and processing station where the data are processed into an image (hardcopy and/or digital).

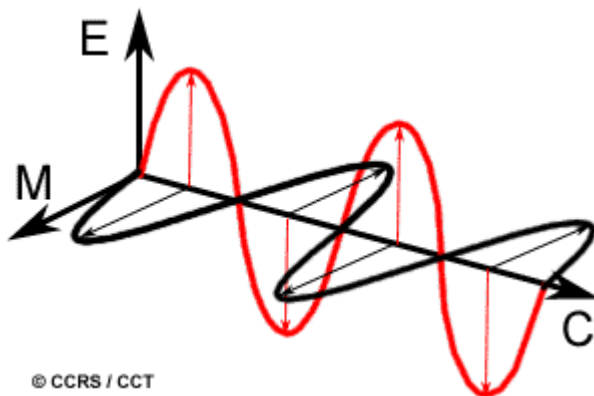
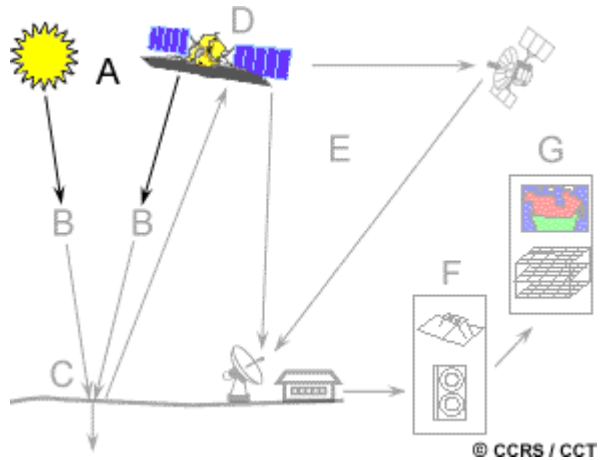
6. **Interpretation and Analysis (F)** - the processed image is interpreted, visually and/or digitally or electronically, to extract information about the target which was illuminated.

7. **Application (G)** - the final element of the remote sensing process is achieved when we apply the information we have been able to extract from the imagery about the target in order to better understand it, reveal some new information, or assist in solving a particular problem.

These seven elements comprise the remote sensing process from beginning to end. We will be covering all of these in sequential order throughout the five chapters of this tutorial, building upon the information learned as we go. Enjoy the journey!

1.2 Electromagnetic Radiation

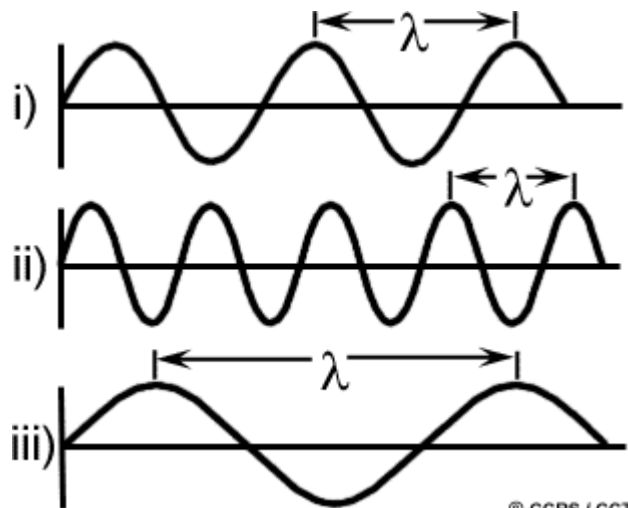
As was noted in the previous section, the first requirement for remote sensing is to have an **energy source to illuminate the target** (unless the sensed energy is being emitted by the target). This energy is in the form of electromagnetic radiation.



All electromagnetic radiation has fundamental properties and behaves in predictable ways according to the basics of wave theory.

Electromagnetic radiation consists of an electrical field (E) which varies in magnitude in a direction perpendicular to the direction in which the radiation is traveling, and a magnetic field (M) oriented at right angles to the electrical field. Both these fields travel at the speed of light (c).

Two characteristics of electromagnetic radiation are particularly important for understanding remote sensing. These are the **wavelength and frequency**.



Section 1.2 Electromagnetic Radiation

The wavelength is the length of one wave cycle, which can be measured as the distance between successive wave crests. Wavelength is usually represented by the Greek letter lambda (λ). Wavelength is measured in metres (m) or some factor of metres such as **nanometres** (nm, 10^{-9} metres), **micrometres** (μm , 10^{-6} metres) (μm , 10^{-6} metres) or centimetres (cm, 10^{-2} metres). Frequency refers to the number of cycles of a wave passing a fixed point per unit of time. Frequency is normally measured in **hertz** (Hz), equivalent to one cycle per second, and various multiples of hertz.

Wavelength and frequency are related by the following formula:

$$c = \lambda \nu$$

where:

λ = wavelength (m)

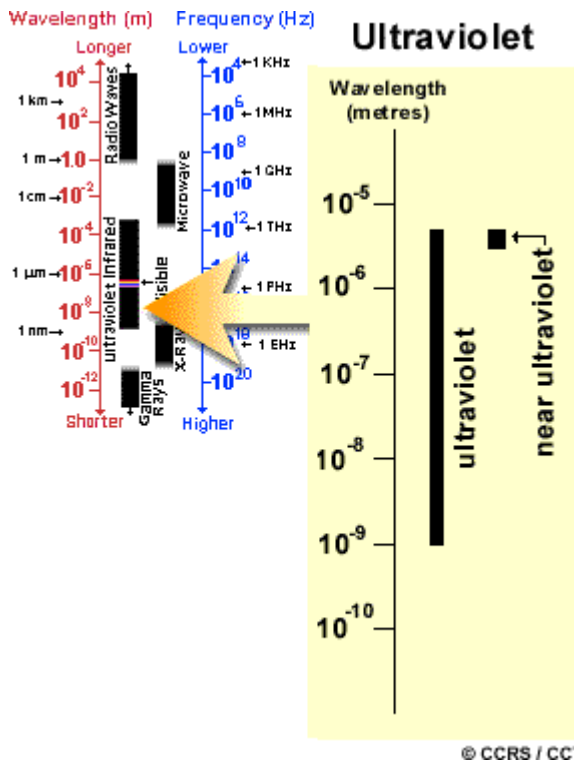
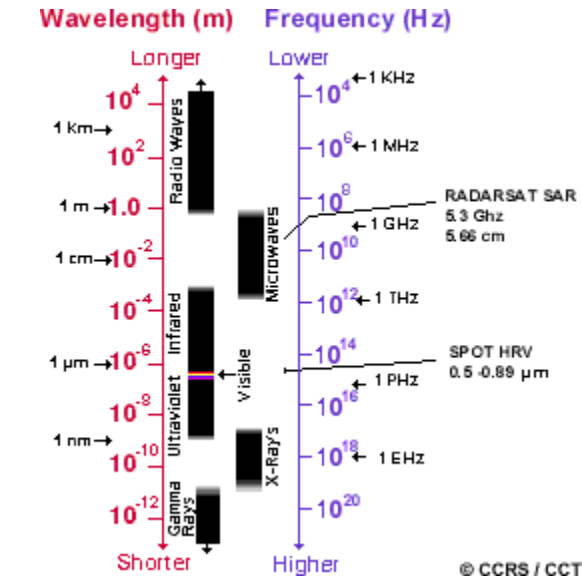
ν = frequency (cycles per second, Hz)

c = speed of light ($3 \times 10^8 \text{ m/s}$)

Therefore, the two are inversely related to each other. The shorter the wavelength, the higher the frequency. The longer the wavelength, the lower the frequency. Understanding the characteristics of electromagnetic radiation in terms of their wavelength and frequency is crucial to understanding the information to be extracted from remote sensing data. Next we will be examining the way in which we categorize electromagnetic radiation for just that purpose.

1.3 The Electromagnetic Spectrum

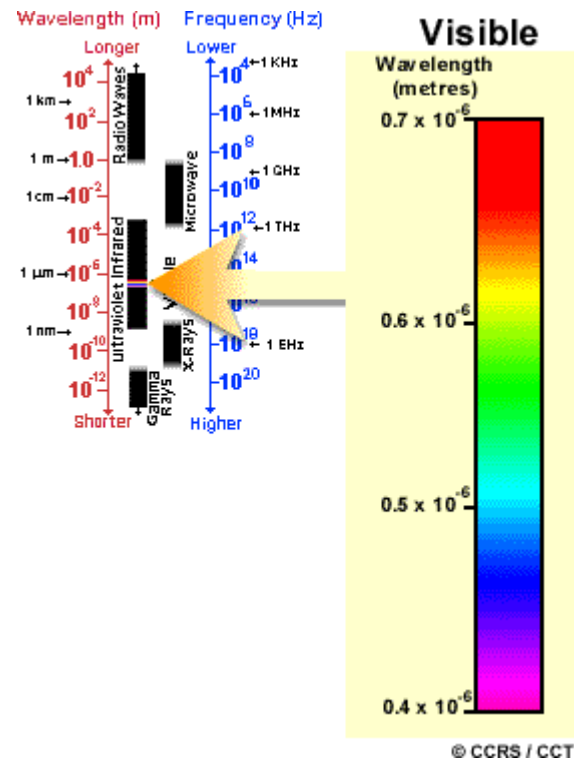
The **electromagnetic spectrum** ranges from the shorter wavelengths (including gamma and x-rays) to the longer wavelengths (including microwaves and broadcast radio waves). There are several regions of the electromagnetic spectrum which are useful for remote sensing.



For most purposes, the **ultraviolet or UV** portion of the spectrum has the shortest wavelengths which are practical for remote sensing. This radiation is just beyond the violet portion of the visible wavelengths, hence its name. Some Earth surface materials, primarily rocks and minerals, fluoresce or emit visible light when illuminated by UV radiation.

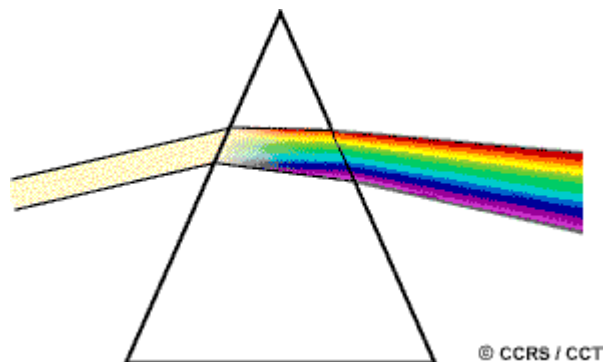
Section 1.3 The Electromagnetic Spectrum

The light which our eyes - our "remote sensors" - can detect is part of the **visible spectrum**. It is important to recognize how small the visible portion is relative to the rest of the spectrum. There is a lot of radiation around us which is "invisible" to our eyes, but can be detected by other remote sensing instruments and used to our advantage. The visible wavelengths cover a range from approximately 0.4 to 0.7 μm . The longest visible wavelength is red and the shortest is violet. Common wavelengths of what we perceive as particular colours from the visible portion of the spectrum are listed below. It is important to note that this is the only portion of the spectrum we can associate with the concept of **colours**.



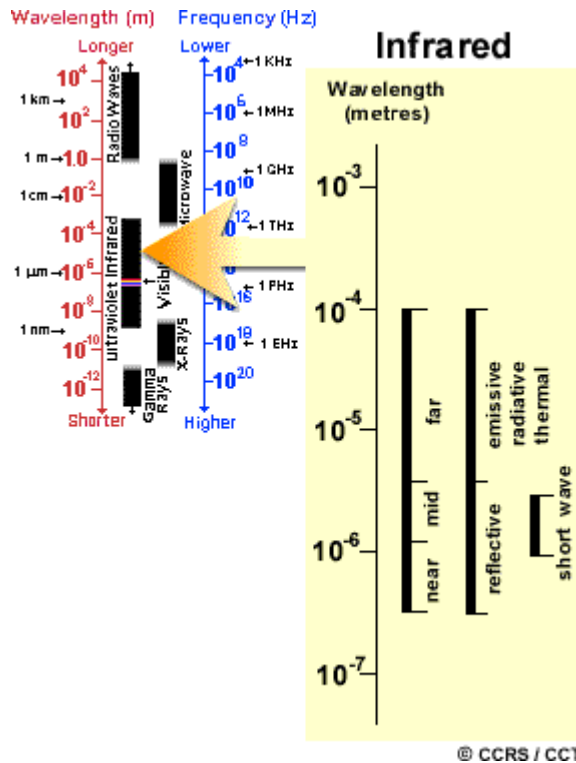
- **Violet:** 0.4 - 0.446 μm
- **Blue:** 0.446 - 0.500 μm
- **Green:** 0.500 - 0.578 μm
- **Yellow:** 0.578 - 0.592 μm
- **Orange:** 0.592 - 0.620 μm
- **Red:** 0.620 - 0.7 μm

Blue, green, and red are the **primary colours** or wavelengths of the visible spectrum. They are defined as such because no single primary colour can be created from the other two, but all other colours can be formed by combining blue, green, and red in various proportions. Although we see sunlight as a uniform or homogeneous colour, it is actually composed of various wavelengths of radiation in primarily the ultraviolet, visible and infrared portions of the spectrum. The visible portion of this radiation can be shown in its



Section 1.3 The Electromagnetic Spectrum

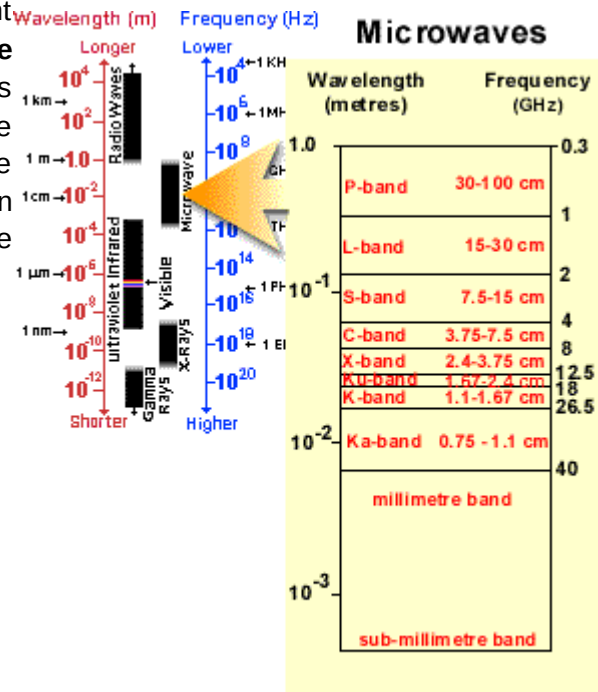
component colours when sunlight is passed through a **prism**, which bends the light in differing amounts according to wavelength.



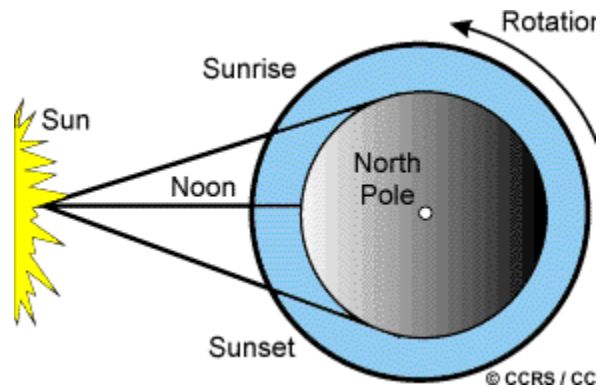
The next portion of the spectrum of interest is the infrared (IR) region which covers the wavelength range from approximately 0.7 μ m to 100 μ m - more than 100 times as wide as the visible portion! The infrared region can be divided into two categories based on their radiation properties - the **reflected IR**, and the emitted or **thermal IR**. Radiation in the reflected IR region is used for remote sensing purposes in ways very similar to radiation in the visible portion. The reflected IR covers wavelengths from approximately 0.7 μ m to 3.0 μ m. The thermal IR region is quite different than the visible and reflected IR portions, as this energy is essentially the radiation that is emitted from the Earth's surface in the form of heat. The thermal IR covers wavelengths from approximately 3.0 μ m to 100 μ m.

The portion of the spectrum of more recent interest to remote sensing is the **microwave region** from about 1 mm to 1 m. This covers the longest wavelengths used for remote sensing. The shorter wavelengths have properties similar to the thermal infrared region while the longer wavelengths approach the wavelengths used for radio broadcasts.

Because of the special nature of this region and its importance to remote sensing in Canada, an entire chapter (Chapter 3) of the tutorial is dedicated to microwave sensing.



Section 1.4 Interactions with the Atmosphere



Rayleigh scattering occurs when particles are very small compared to the wavelength of the radiation. These could be particles such as small specks of dust or nitrogen and oxygen molecules. Rayleigh scattering causes shorter wavelengths of energy to be scattered much more than longer wavelengths. Rayleigh scattering is the dominant scattering mechanism in the upper atmosphere. The fact that the sky appears "blue" during the day is because of this phenomenon. As sunlight passes through the atmosphere, the shorter wavelengths (i.e. blue) of the visible spectrum are scattered more than the other (longer) visible wavelengths. At **sunrise and sunset** the light has to travel farther through the atmosphere than at midday and the scattering of the shorter wavelengths is more complete; this leaves a greater proportion of the longer wavelengths to penetrate the atmosphere.

Mie scattering occurs when the particles are just about the same size as the wavelength of the radiation. Dust, pollen, smoke and water vapour are common causes of Mie scattering which tends to affect longer wavelengths than those affected by Rayleigh scattering. Mie scattering occurs mostly in the lower portions of the atmosphere where larger particles are more abundant, and dominates when cloud conditions are overcast.



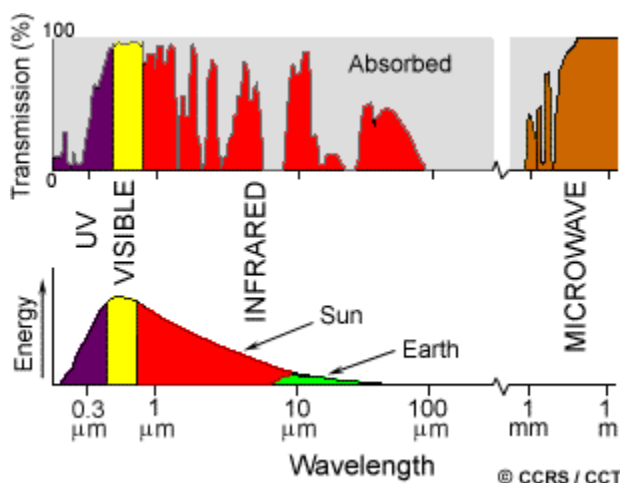
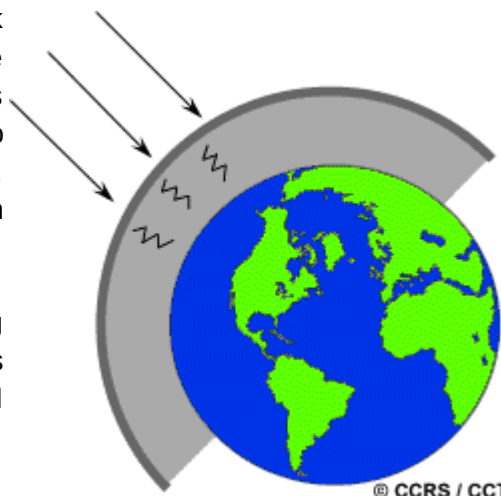
The final scattering mechanism of importance is called **nonselective scattering**. This occurs when the particles are much larger than the wavelength of the radiation. Water droplets and large dust particles can cause this type of scattering. Nonselective scattering gets its name from the fact that all wavelengths are scattered about equally. This type of scattering causes fog and clouds to appear white to our eyes because blue, green, and red light are all scattered in approximately equal quantities (blue+green+red light = white light).

Section 1.4 Interactions with the Atmosphere

Absorption is the other main mechanism at work when electromagnetic radiation interacts with the atmosphere. In contrast to scattering, this phenomenon causes molecules in the atmosphere to absorb energy at various wavelengths. Ozone, carbon dioxide, and water vapour are the three main atmospheric constituents which absorb radiation.

Ozone serves to absorb the harmful (to most living things) ultraviolet radiation from the sun. Without this protective layer in the atmosphere our skin would burn when exposed to sunlight.

You may have heard **carbon dioxide** referred to as a greenhouse gas. This is because it tends to absorb radiation strongly in the far infrared portion of the spectrum - that area associated with thermal heating - which serves to trap this heat inside the atmosphere. Water vapour in the atmosphere absorbs much of the incoming longwave infrared and shortwave microwave radiation (between $22\mu\text{m}$ and 1m). The presence of water vapour in the lower atmosphere varies greatly from location to location and at different times of the year. For example, the air mass above a desert would have very little water vapour to absorb energy, while the tropics would have high concentrations of water vapour (i.e. high humidity).



Because these gases absorb electromagnetic energy in very specific regions of the spectrum, they influence where (in the spectrum) we can "look" for remote sensing purposes. Those areas of the spectrum which are not severely influenced by atmospheric absorption and thus, are useful to remote sensors, are called **atmospheric windows**. By comparing the characteristics of the two most common energy/radiation sources (the sun and the earth) with the

atmospheric windows available to us, we can define those wavelengths that we can use **most effectively** for remote sensing. The visible portion of the spectrum, to which our eyes are most sensitive, corresponds to both an atmospheric window and the peak energy level of the sun. Note also that heat energy emitted by the Earth corresponds to a window around $10\mu\text{m}$ in the thermal IR portion of the spectrum, while the large window at wavelengths beyond 1mm is associated with the

Section 1.4 Interactions with the Atmosphere

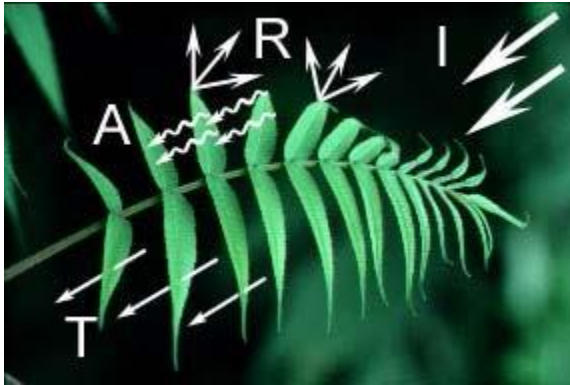
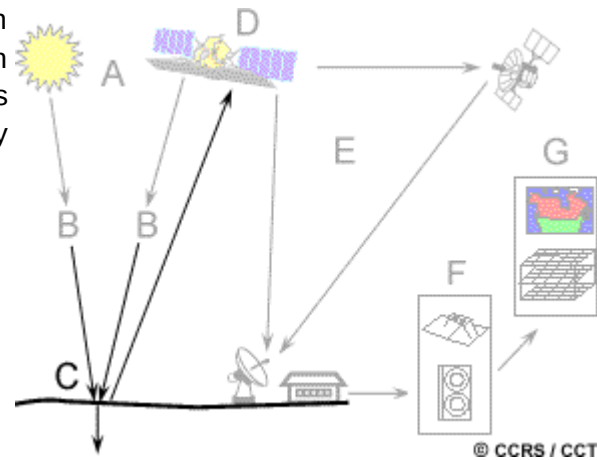
microwave region.

Now that we understand how electromagnetic energy makes its journey from its source to the surface (and it is a difficult journey, as you can see) we will next examine what happens to that radiation when it does arrive at the Earth's surface.

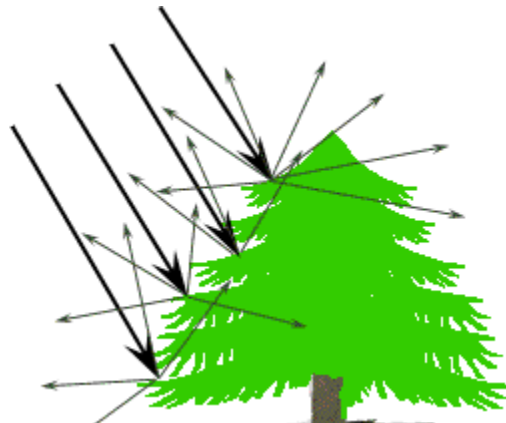
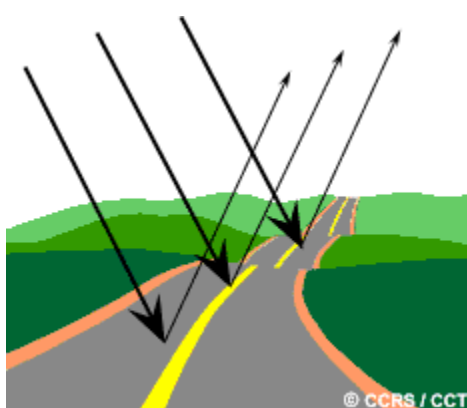
1.5 Radiation - Target Interactions

Radiation that is not absorbed or scattered in the atmosphere can reach and interact with the Earth's surface. There are three (3) forms of interaction that can take place when energy strikes, or is **incident (I)** upon the surface.

These are: **absorption (A)**; **transmission (T)**; and **reflection (R)**. The total incident energy will interact with the surface in one or more of these three ways. The proportions of each will depend on the wavelength of the energy and the material and condition of the feature.



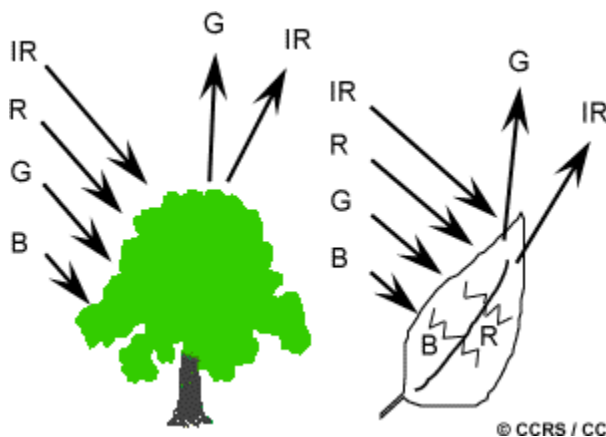
Absorption (A) occurs when radiation (energy) is absorbed into the target while transmission (T) occurs when radiation passes through a target. Reflection (R) occurs when radiation "bounces" off the target and is redirected. In remote sensing, we are most interested in measuring the radiation reflected from targets. We refer to two types of reflection, which represent the two extreme ends of the way in which energy is reflected from a target: **specular reflection** and **diffuse reflection**.



Section 1.5 Radiation - Target Interactions

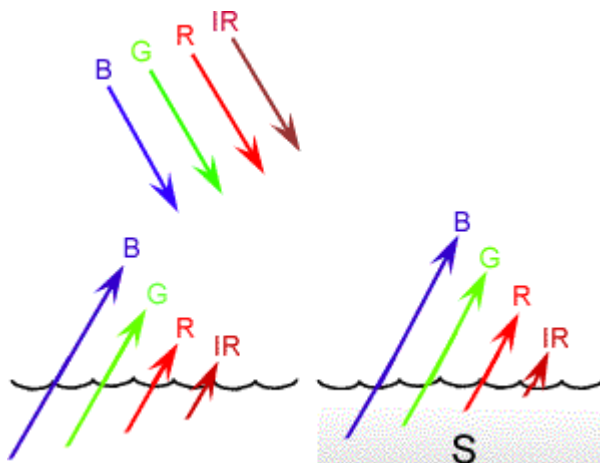
When a surface is smooth we get **specular** or mirror-like reflection where all (or almost all) of the energy is directed away from the surface in a single direction. **Diffuse** reflection occurs when the surface is rough and the energy is reflected almost uniformly in all directions. Most earth surface features lie somewhere between perfectly specular or perfectly diffuse reflectors. Whether a particular target reflects specularly or diffusely, or somewhere in between, depends on the surface roughness of the feature in comparison to the wavelength of the incoming radiation. If the wavelengths are much smaller than the surface variations or the particle sizes that make up the surface, diffuse reflection will dominate. For example, fine-grained sand would appear fairly smooth to long wavelength microwaves but would appear quite rough to the visible wavelengths.

Let's take a look at a couple of examples of targets at the Earth's surface and how energy at the visible and infrared wavelengths interacts with them.



Leaves: A chemical compound in leaves called chlorophyll strongly absorbs radiation in the red and blue wavelengths but reflects green wavelengths. Leaves appear "greenest" to us in the summer, when chlorophyll content is at its maximum. In autumn, there is less chlorophyll in the leaves, so there is less absorption and proportionately more reflection of the red wavelengths, making the leaves appear red or yellow (yellow is a combination of red and green wavelengths). The

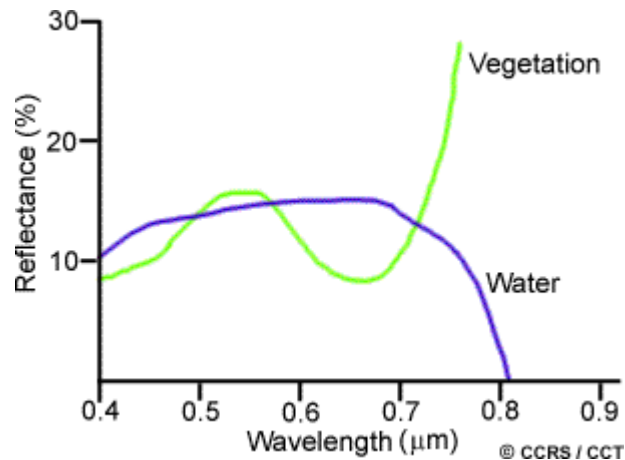
internal structure of healthy leaves act as excellent diffuse reflectors of near-infrared wavelengths. If our eyes were sensitive to near-infrared, trees would appear extremely bright to us at these wavelengths. In fact, measuring and monitoring the near-IR reflectance is one way that scientists can determine how healthy (or unhealthy) vegetation may be.



Water: Longer wavelength visible and near infrared radiation is absorbed more by water than shorter visible wavelengths. Thus water typically looks blue or blue-green due to stronger reflectance at these shorter wavelengths, and darker if viewed at red or near infrared wavelengths. If there is suspended sediment present in the upper layers of the water body, then this will allow better reflectivity and a brighter appearance of the water. The apparent colour of the water will show a slight shift to longer

Section 1.5 Radiation - Target Interactions

wavelengths. Suspended sediment (S) can be easily confused with shallow (but clear) water, since these two phenomena appear very similar. Chlorophyll in algae absorbs more of the blue wavelengths and reflects the green, making the water appear more green in colour when algae is present. The topography of the water surface (rough, smooth, floating materials, etc.) can also lead to complications for water-related interpretation due to potential problems of specular reflection and other influences on colour and brightness.

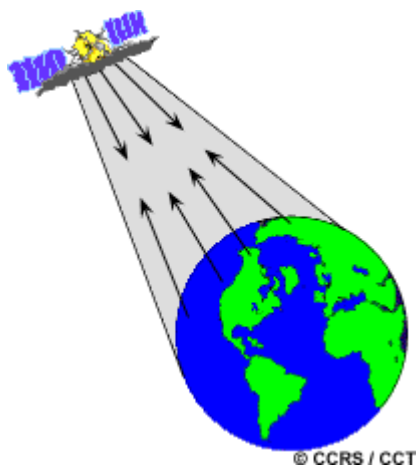
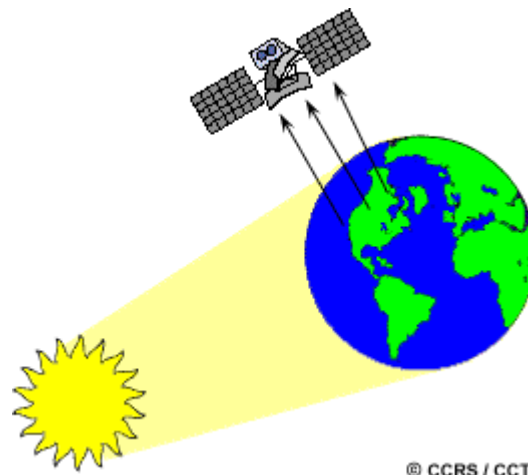


We can see from these examples that, depending on the complex make-up of the target that is being looked at, and the wavelengths of radiation involved, we can observe very different responses to the mechanisms of absorption, transmission, and reflection. By measuring the energy that is reflected (or emitted) by targets on the Earth's surface over a variety of different wavelengths, we can build up a **spectral response** for that object. By comparing the response patterns of different features we may be able to distinguish between them, where we might not be able to, if we only compared them at one wavelength. For example, water and vegetation may reflect somewhat similarly in the visible wavelengths but are almost always separable in the infrared. Spectral response can be quite variable, even for the same target type, and can also vary with time (e.g. "green-ness" of leaves) and location. Knowing where to "look" spectrally and understanding the factors which influence the spectral response of the features of interest are critical to correctly interpreting the interaction of electromagnetic radiation with the surface.

1.6 Passive vs. Active Sensing

So far, throughout this chapter, we have made various references to the sun as a source of energy or radiation. The sun provides a very convenient source of energy for remote sensing. The sun's energy is either **reflected**, as it is for visible wavelengths, or absorbed and then **re-emitted**, as it is for thermal infrared wavelengths. Remote sensing systems which measure energy that is naturally available are called **passive sensors**. Passive sensors can only be used to detect energy when the naturally occurring energy is available. For all reflected energy, this can only take place during the time when the sun is illuminating the Earth. There is

no reflected energy available from the sun at night. Energy that is naturally emitted (such as thermal infrared) can be detected day or night, as long as the amount of energy is large enough to be recorded.



Active sensors, on the other hand, provide their own energy source for illumination. The sensor emits radiation which is directed toward the target to be investigated. The radiation reflected from that target is detected and measured by the sensor. Advantages for active sensors include the ability to obtain measurements anytime, regardless of the time of day or season. Active sensors can be used for examining wavelengths that are not sufficiently provided by the sun, such as microwaves, or to better control the way a target is illuminated. However, active systems require the generation of a fairly large amount of energy to adequately illuminate targets. Some examples of active sensors are a laser fluorosensor and a synthetic

aperture radar (SAR).

1.7 Characteristics of Images

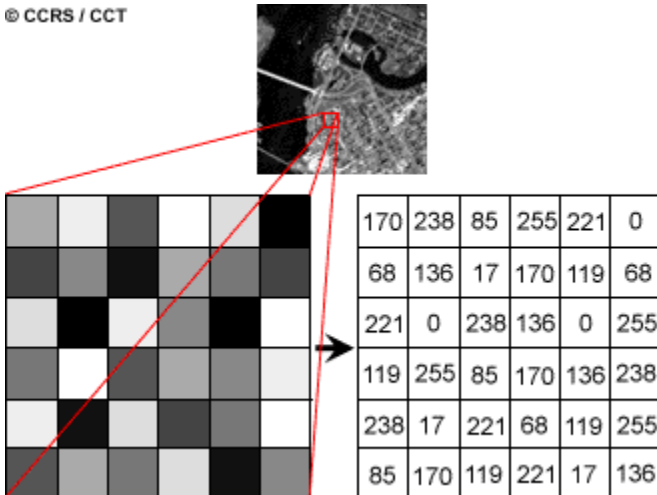
Before we go on to the next chapter, which looks in more detail at sensors and their characteristics, we need to define and understand a few fundamental terms and concepts associated with remote sensing images.



Electromagnetic energy may be detected either photographically or electronically. The photographic process uses chemical reactions on the surface of light-sensitive film to detect and record energy variations. It is important to distinguish between the terms **images** and **photographs** in remote sensing. An image refers to any pictorial representation, regardless of what wavelengths or remote sensing device has been used to detect and record the electromagnetic energy. A **photograph** refers

specifically to images that have been detected as well as recorded on photographic film. The black and white photo to the left, of part of the city of Ottawa, Canada was taken in the visible part of the spectrum. Photos are normally recorded over the wavelength range from $0.3\ \mu\text{m}$ to $0.9\ \mu\text{m}$ - the visible and reflected infrared. Based on these definitions, we can say that all photographs are images, but not all images are photographs. Therefore, unless we are talking specifically about an image recorded photographically, we use the term image.

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A photograph could also be represented and displayed in a **digital** format by subdividing the image into small equal-sized and shaped areas, called picture elements or **pixels**, and representing the brightness of each area with a numeric value or **digital number**. Indeed, that is exactly what has been done to the photo to the left. In fact, using the definitions we have just discussed, this is actually a **digital image** of the

original photograph! The photograph was scanned and subdivided into pixels with each pixel assigned a digital number representing its relative brightness. The computer displays each digital value as different brightness levels. Sensors that

Section 1.7 Characteristics of Images

record electromagnetic energy, electronically record the energy as an array of numbers in digital format right from the start. These two different ways of representing and displaying remote sensing data, either pictorially or digitally, are interchangeable as they convey the same information (although some detail may be lost when converting back and forth).

In previous sections we described the visible portion of the spectrum and the concept of colours. We see colour because our eyes detect the entire visible range of wavelengths and our brains process the information into separate colours. Can you imagine what the world would look like if we could only see very narrow ranges of wavelengths or colours? That is how many sensors work. The information from a narrow wavelength range is gathered and stored in a **channel**, also sometimes referred to as a **band**. We can combine and display channels of information digitally using the three primary colours (blue, green, and red). The data from each channel is represented as one of the primary colours and, depending on the relative brightness (i.e. the digital value) of each pixel in each channel, the primary colours combine in different proportions to represent different colours.



When we use this method to display a single channel or range of wavelengths, we are actually displaying that channel through all three primary colours. Because the brightness level of each pixel is the same for each primary colour, they combine to form a **black and white image**, showing various shades of gray from black to white. When we display more than one channel each as a different primary colour, then the brightness levels may be different for each channel/primary colour combination and they will combine to form a **colour image**.

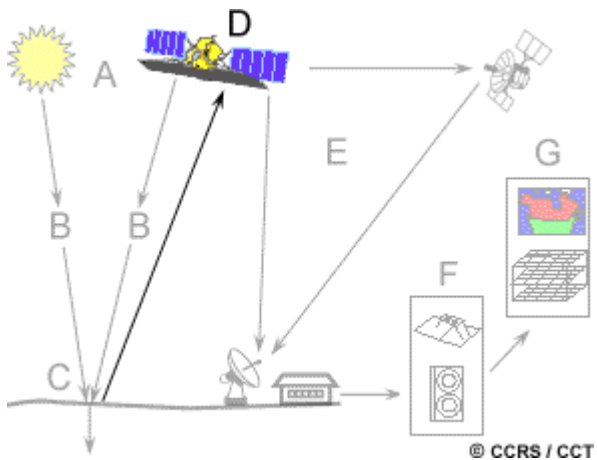


2. Satellites and Sensors

1.1 On the Ground, In the Air, In Space

In Chapter 1 we learned some of the fundamental concepts required to understand the process that encompasses remote sensing. We covered in some detail the first three components of this process: the energy source, interaction of energy with the atmosphere, and interaction of energy with the surface. We touched briefly on the fourth component - **recording of energy by the sensor** - when we discussed passive vs. active sensors and characteristics of images. In this chapter, we will take a closer look at this component of the remote sensing process

by examining in greater detail, the characteristics of remote sensing platforms and sensors and the data they collect. We will also touch briefly on how those data are processed once they have been recorded by the sensor.



In order for a sensor to collect and record energy reflected or emitted from a target or surface, it must reside on a stable **platform** removed from the target or surface being observed. Platforms for remote sensors may be situated on the ground, on an aircraft or balloon (or some other platform within the Earth's atmosphere), or on a spacecraft or satellite outside of the Earth's atmosphere.

Ground-based sensors are often used to record detailed information about the surface which is compared with information collected from aircraft or satellite sensors. In some cases,

Section 2.1 On the Ground, In the Air, In Space

this can be used to better characterize the target which is being imaged by these other sensors, making it possible to better understand the information in the imagery.



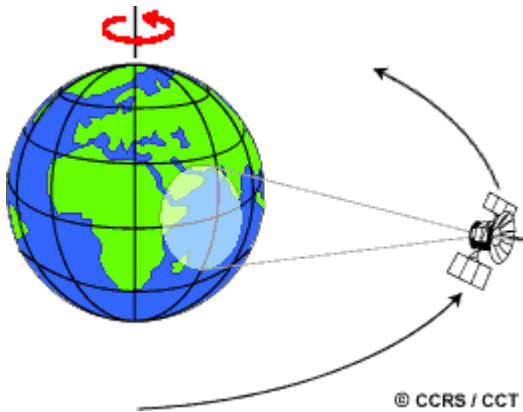
Sensors may be placed on a ladder, scaffolding, tall building, cherry-picker, crane, etc. Aerial platforms are primarily stable wing **aircraft**, although helicopters are occasionally used. Aircraft are often used to collect very detailed images and facilitate the collection of data over virtually any portion of the Earth's surface at any time.

In space, remote sensing is sometimes conducted from the **space shuttle** or, more commonly, from satellites. **Satellites** are objects which revolve around another object - in this case, the Earth. For example, the moon is a natural satellite, whereas man-made satellites include those platforms launched for remote sensing, communication, and telemetry (location and navigation) purposes. Because of their orbits, satellites permit repetitive coverage of the Earth's surface on a continuing basis. Cost is often a significant factor in choosing among the various platform options.



1.2 Satellite Characteristics: Orbits and Swaths

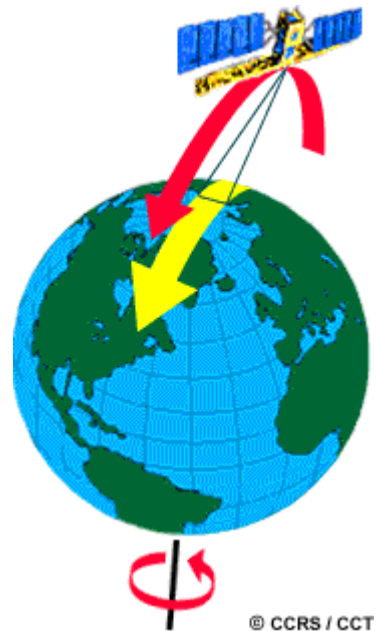
We learned in the previous section that remote sensing instruments can be placed on a variety of platforms to view and image targets. Although ground-based and aircraft platforms may be used, satellites provide a great deal of the remote sensing imagery commonly used today. Satellites have several unique characteristics which make them particularly useful for remote sensing of the Earth's surface.



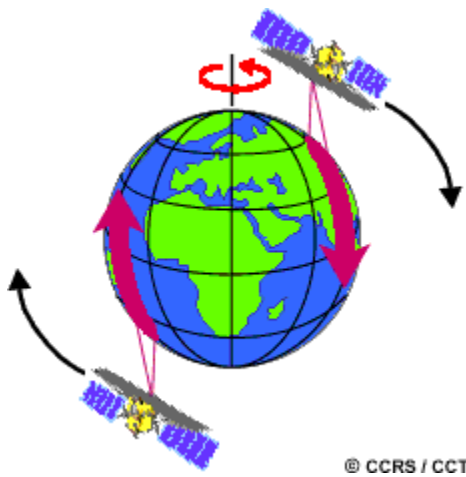
The path followed by a satellite is referred to as its **orbit**. Satellite orbits are matched to the capability and objective of the sensor(s) they carry. Orbit selection can vary in terms of altitude (their height above the Earth's surface) and their orientation and rotation relative to the Earth. Satellites at very high altitudes, which view the same portion of the Earth's surface at all times have **geostationary orbits**. These geostationary satellites, at altitudes of approximately 36,000 kilometres, revolve at speeds which match the rotation of the Earth so they seem stationary, relative to the Earth's

surface. This allows the satellites to observe and collect information continuously over specific areas. Weather and communications satellites commonly have these types of orbits. Due to their high altitude, some geostationary weather satellites can monitor weather and cloud patterns covering an entire hemisphere of the Earth.

Many remote sensing platforms are designed to follow an orbit (basically north-south) which, in conjunction with the Earth's rotation (west-east), allows them to cover most of the Earth's surface over a certain period of time. These are **near-polar orbits**, so named for the inclination of the orbit relative to a line running between the North and South poles. Many of these satellite orbits are also **sun-synchronous** such that they cover each area of the world at a constant local time of day called **local sun time**. At any given latitude, the position of the sun in the sky as the satellite passes overhead will be the same within the same season. This ensures consistent illumination conditions when acquiring images in a specific season over successive years, or over a particular area over a series of days. This is an important factor for monitoring changes between images or for mosaicking adjacent images together, as they do not have to be corrected for different illumination conditions.

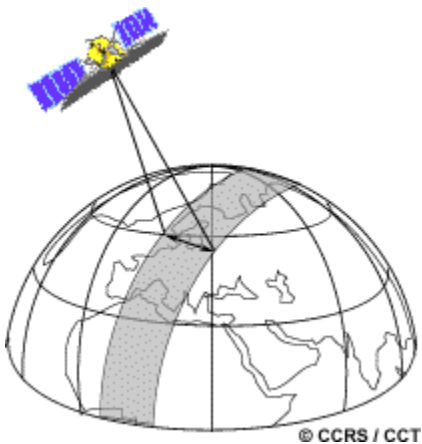


Section 2.2 Satellite Characteristics: Orbits and Swaths



Most of the remote sensing satellite platforms today are in near-polar orbits, which means that the satellite travels northwards on one side of the Earth and then toward the southern pole on the second half of its orbit. These are called **ascending and descending passes**, respectively. If the orbit is also sun-synchronous, the ascending pass is most likely on the shadowed side of the Earth while the descending pass is on the sunlit side. Sensors recording reflected solar energy only image the surface on a descending pass, when solar illumination is available. Active sensors which provide their own illumination or passive sensors that record emitted (e.g. thermal) radiation can also image the surface on ascending

passes.



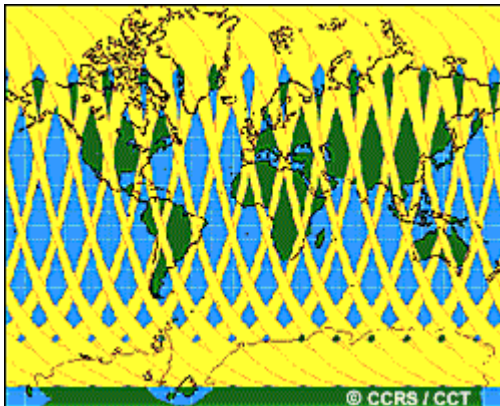
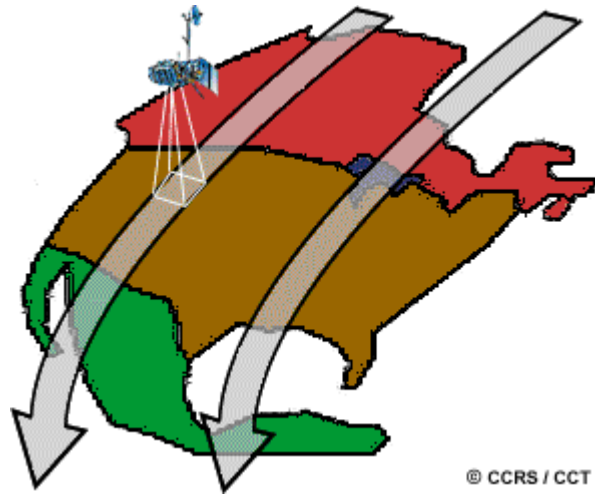
As a satellite revolves around the Earth, the sensor "sees" a certain portion of the Earth's surface. The area imaged on the surface, is referred to as the **swath**. Imaging swaths for spaceborne sensors generally vary between tens and hundreds of kilometres wide. As the satellite orbits the Earth from pole to pole, its east-west position wouldn't change if the Earth didn't rotate. However, as seen from the Earth, it seems that the satellite is shifting westward because the Earth is rotating (from west to east) beneath it. This apparent movement allows the satellite swath to cover a **new area with each consecutive pass**. The satellite's orbit and the rotation of the Earth work together to allow complete coverage of

the Earth's surface, after it has completed one complete cycle of orbits.

Section 2.2 Satellite Characteristics: Orbits and Swaths

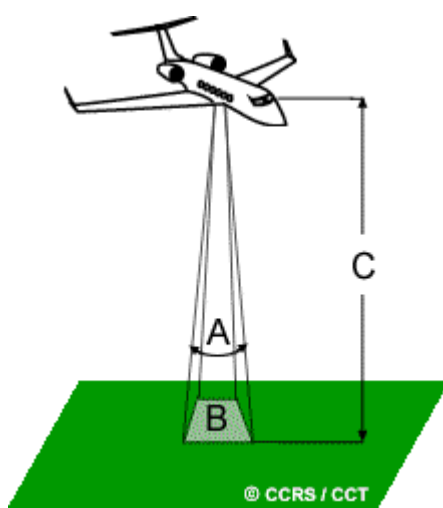
If we start with any randomly selected pass in a satellite's orbit, an orbit cycle will be completed when the satellite retraces its path, passing over the same point on the Earth's surface directly below the satellite (called the **nadir** point) for a second time. The exact length of time of the orbital cycle will vary with each satellite. The interval of time required for the satellite to complete its orbit cycle is not the same as the "**revisit period**". Using steerable sensors, an

satellite-borne instrument can view an area (off-nadir) before and after the orbit passes over a target, thus making the 'revisit' time less than the orbit cycle time. The revisit period is an important consideration for a number of monitoring applications, especially when frequent imaging is required (for example, to monitor the spread of an oil spill, or the extent of flooding). In near-polar orbits, areas at high latitudes will be imaged more frequently than the equatorial zone due to the increasing **overlap in adjacent swaths** as the orbit paths come closer together near the poles.



1.3 Spatial Resolution, Pixel Size, and Scale

For some remote sensing instruments, the distance between the target being imaged and the platform, plays a large role in determining the detail of information obtained and the total area imaged by the sensor. Sensors onboard platforms far away from their targets, typically view a larger area, but cannot provide great detail. Compare what an astronaut onboard the space shuttle sees of the Earth to what you can see from an airplane. The astronaut might see your whole province or country in one glance, but couldn't distinguish individual houses. Flying over a city or town, you would be able to see individual buildings and cars, but you would be viewing a much smaller area than the astronaut. There is a similar difference between satellite images and airphotos.



The detail discernible in an image is dependent on the **spatial resolution** of the sensor and refers to the size of the smallest possible feature that can be detected.

Spatial resolution of passive sensors (we will look at the special case of active microwave sensors later) depends primarily on their **Instantaneous Field of View (IFOV)**. The IFOV is the angular cone of visibility of the sensor (A) and determines the area on the Earth's surface which is "seen" from a given altitude at one particular moment in time (B). The size of the area viewed is determined by multiplying the IFOV by the distance from the ground to the sensor (C). This area on the ground is called the **resolution cell** and determines a sensor's maximum spatial resolution. For a homogeneous feature to be

detected, its size generally has to be equal to or larger than the resolution cell. If the feature is smaller than this, it may not be detectable as the average brightness of all features in that resolution cell will be recorded. However, smaller features may sometimes be detectable if their reflectance dominates within a particular resolution cell allowing sub-pixel or resolution cell detection.

As we mentioned in Chapter 1, most remote sensing images are composed of a matrix of picture elements, or **pixels**, which are the smallest units of an image. Image pixels are normally square and represent a certain area on an image. It is important to distinguish between pixel size and spatial resolution - they are not interchangeable. If a sensor has a spatial resolution of 20 metres and an image from that sensor is displayed at full resolution, each pixel represents an area of 20m x 20m on the ground. In this case the pixel size and resolution are the same. However, it is possible to display an image with a pixel size different than the resolution. Many posters of satellite images of the Earth have their pixels averaged to represent larger areas, although the original spatial resolution of the sensor that collected the imagery remains the same.

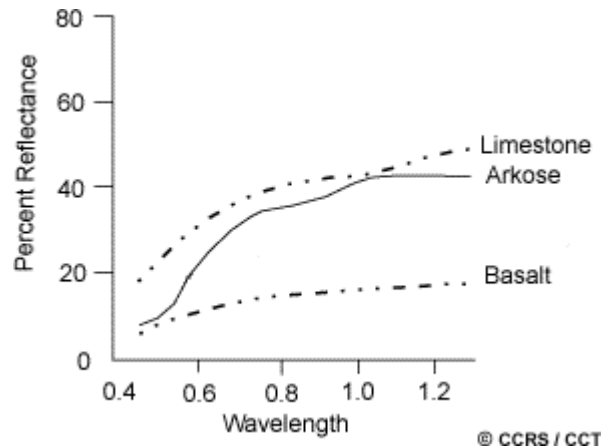
Section 2.3 Spatial Resolution, Pixel Size, and Scale



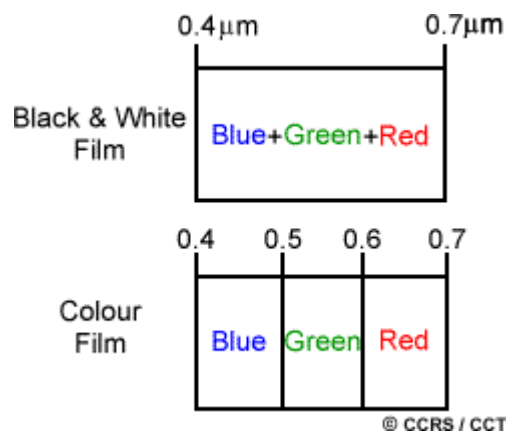
Images where only large features are visible are said to have **coarse or low resolution**. In **fine or high resolution** images, small objects can be detected. Military sensors for example, are designed to view as much detail as possible, and therefore have very fine resolution. Commercial satellites provide imagery with resolutions varying from a few metres to several kilometres. Generally speaking, the finer the resolution, the less total ground area can be seen.

The ratio of distance on an image or map, to actual ground distance is referred to as scale. If you had a map with a scale of 1:100,000, an object of 1cm length on the map would actually be an object 100,000cm (1km) long on the ground. Maps or images with small "map-to-ground ratios" are referred to as small scale (e.g. 1:100,000), and those with larger ratios (e.g. 1:5,000) are called large scale.

1.4 Spectral Resolution



In Chapter 1, we learned about **spectral response** and **spectral emissivity curves** which characterize the reflectance and/or emittance of a feature or target over a variety of wavelengths. Different classes of features and details in an image can often be distinguished by comparing their responses over distinct wavelength ranges. Broad classes, such as water and vegetation, can usually be separated using very broad wavelength ranges - the visible and near infrared - as we learned in section 1.5. Other more specific classes, such as **different rock types**, may not be easily distinguishable using either of these broad wavelength ranges and would require comparison at much finer wavelength ranges to separate them. Thus, we would require a sensor with higher **spectral resolution**. Spectral resolution describes the ability of a sensor to define fine wavelength intervals. The finer the spectral resolution, the narrower the wavelength range for a particular channel or band.



Black and white film records wavelengths extending over much, or all of the visible portion of the electromagnetic spectrum. Its **spectral resolution** is fairly coarse, as the various wavelengths of the visible spectrum are not individually distinguished and the overall

Section 2.4 Spectral Resolution

reflectance in the entire visible portion is recorded. Colour film is also sensitive to the reflected energy over the visible portion of the spectrum, but has higher spectral resolution, as it is individually sensitive to the reflected energy at the blue, green, and red wavelengths of the spectrum. Thus, it can represent features of various colours based on their reflectance in each of these distinct wavelength ranges.

Many remote sensing systems record energy over several separate wavelength ranges at various spectral resolutions. These are referred to as **multi-spectral sensors** and will be described in some detail in following sections. Advanced multi-spectral sensors called **hyperspectral** sensors, detect hundreds of very narrow spectral bands throughout the visible, near-infrared, and mid-infrared portions of the electromagnetic spectrum. Their very high spectral resolution facilitates fine discrimination between different targets based on their spectral response in each of the narrow bands.

1.5 Radiometric Resolution

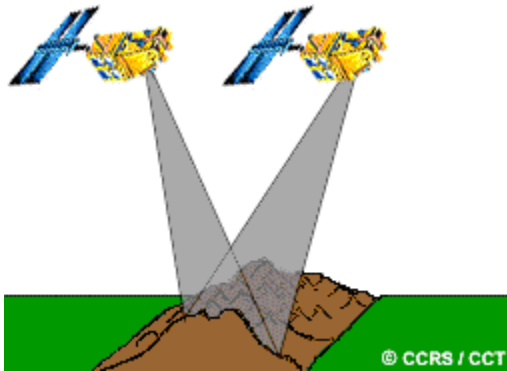
While the arrangement of pixels describes the spatial structure of an image, the radiometric characteristics describe the actual information content in an image. Every time an image is acquired on film or by a sensor, its sensitivity to the magnitude of the electromagnetic energy determines the **radiometric resolution**. The radiometric resolution of an imaging system describes its ability to discriminate very slight differences in energy. The finer the radiometric resolution of a sensor, the more sensitive it is to detecting small differences in reflected or emitted energy.



Imagery data are represented by positive digital numbers which vary from 0 to (one less than) a selected power of 2. This range corresponds to the number of bits used for coding numbers in binary format. Each bit records an exponent of power 2 (e.g. 1 bit = $2^1 = 2$). The maximum number of brightness levels available depends on the number of bits used in representing the energy recorded. Thus, if a sensor used 8 bits to record the data, there would be $2^8 = 256$ digital values available, ranging from 0 to 255. However, if only 4 bits were used, then only $2^4 = 16$ values ranging from 0 to 15 would be available. Thus, the radiometric resolution would be much less. Image data are generally displayed in a range of grey tones, with black representing a digital number of 0 and white representing the maximum value (for example, 255 in 8-bit data). By **comparing a 2-bit image with an 8-bit image**, we can see that there is a large difference in the level of detail discernible depending on their radiometric resolutions.

1.6 Temporal Resolution

In addition to spatial, spectral, and radiometric resolution, the concept of **temporal resolution** is also important to consider in a remote sensing system. We alluded to this idea in section 2.2 when we discussed the concept of revisit period, which refers to the length of time it takes for a satellite to complete one entire orbit cycle. The revisit period of a satellite sensor is usually several days. Therefore the absolute temporal resolution of a remote sensing system to image the exact same area at the same viewing angle a second time is equal to this period. However, because of some degree of overlap in the imaging swaths of adjacent orbits for most satellites and the increase in this overlap with increasing latitude, some areas of the Earth tend to be re-imaged more frequently. Also, some satellite systems are able to **point their sensors to image the same area** between different satellite passes separated by periods from one to five days. Thus, the actual temporal resolution of a sensor depends on a variety of factors, including the satellite/sensor capabilities, the swath overlap, and latitude.

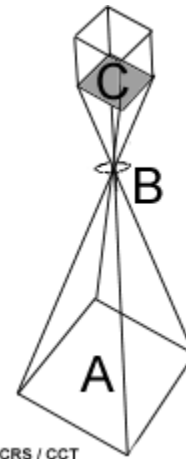


The ability to collect imagery of the same area of the Earth's surface at different periods of time is one of the most important elements for applying remote sensing data. Spectral characteristics of features may change over time and these changes can be detected by collecting and comparing **multi-temporal** imagery. For example, during the growing season, most species of vegetation are in a continual state of change and our ability to monitor those subtle changes using remote sensing is dependent on when and how frequently we collect imagery. By imaging on a continuing basis at different times we are able to monitor the changes that take place on the Earth's surface, whether they are naturally occurring (such as changes in natural vegetation cover or flooding) or induced by humans (such as urban development or deforestation). The time factor in imaging is important when:

- persistent clouds offer limited clear views of the Earth's surface (often in the tropics)
- short-lived phenomena (floods, oil slicks, etc.) need to be imaged
- multi-temporal comparisons are required (e.g. the spread of a forest disease from one year to the next)
- the changing appearance of a feature over time can be used to distinguish it from near-similar features (wheat / maize)

2.7 Cameras and Aerial Photography

Cameras and their use for aerial photography are the simplest and oldest of sensors used for remote sensing of the Earth's surface. Cameras are **framing systems** which acquire a near-instantaneous "snapshot" of an **area (A)**, of the surface. Camera systems are passive optical sensors that use a **lens (B)** (or system of lenses collectively referred to as the optics) to form an image at the focal plane (C), the plane at which an image is sharply defined.



Photographic films are sensitive to light from $0.3\ \mu\text{m}$ to $0.9\ \mu\text{m}$ in wavelength covering the ultraviolet (UV), visible, and near-infrared (NIR). **Panchromatic** films are sensitive to the UV and the visible portions of the spectrum. Panchromatic film produces black and white images and is the most common type of film used for aerial photography. UV photography also uses panchromatic film, but a filter is used with the camera to absorb and block the visible energy from reaching the film. As a result, only the UV reflectance from targets is recorded. UV photography is not widely used, because of the atmospheric scattering and absorption that occurs in this region of the spectrum. Black and white infrared photography uses film sensitive to the entire 0.3 to $0.9\ \mu\text{m}$ wavelength range and is useful for detecting differences in vegetation cover, due to its sensitivity to IR reflectance.



Colour and false colour (or colour infrared, CIR) photography involves the use of a three layer film with each layer sensitive to different ranges of light. For a **normal colour photograph**, the layers are sensitive to blue, green, and red light - the same as our eyes. These photos appear to us the same way that our eyes see the environment, as the colours resemble those which would appear to us as "normal" (i.e. trees appear green, etc.). In colour infrared (CIR) photography, the three emulsion layers are sensitive to green, red, and the photographic portion of near-infrared radiation, which are processed to appear as blue, green, and red,

Section 2.7 Cameras and Aerial Photography

respectively. In a **false colour photograph**, targets with high near-infrared reflectance appear red, those with a high red reflectance appear green, and those with a high green reflectance appear blue, thus giving us a "false" presentation of the targets relative to the colour we normally perceive them to be.

Cameras can be used on a variety of platforms including ground-based stages, helicopters, aircraft, and spacecraft. Very detailed photographs taken from aircraft are useful for many applications where identification of detail or small targets is required. The ground coverage of a photo depends on several factors, including the focal length of the lens, the platform altitude, and the format and size of the film. The focal length effectively controls the **angular field of view** of the lens (similar to the concept of instantaneous field of view discussed in section 2.3) and determines the area "seen" by the camera. Typical focal lengths used are 90mm, 210mm, and most commonly, 152mm. The longer the focal length, the smaller the area covered on the ground, but with greater detail (i.e. larger scale). The area covered also depends on the altitude of the platform. At high altitudes, a camera will "see" a larger area on the ground than at lower altitudes, but with reduced detail (i.e. smaller scale). Aerial photos can provide fine detail down to spatial resolutions of less than 50 cm. A photo's exact spatial resolution varies as a complex function of many factors which vary with each acquisition of data.

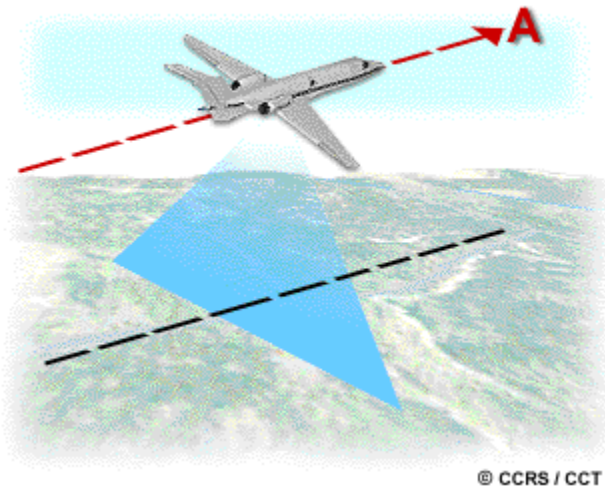


Most aerial photographs are classified as either **oblique** or **vertical**, depending on the orientation of the camera relative to the ground during acquisition. **Oblique aerial photographs** are taken with the camera pointed to the side of the aircraft. High oblique photographs usually include the horizon while low oblique photographs do not. Oblique photographs can be useful for covering very large areas in a single image and for depicting terrain relief and scale. However, they are not widely used for mapping as distortions in scale from the foreground to the background preclude easy measurements of distance, area, and elevation.

Vertical photographs taken with a single-lens frame camera is the most common use of aerial photography for remote sensing and mapping purposes. These cameras are specifically built for capturing a rapid sequence of photographs while limiting geometric distortion. They are often linked with navigation systems onboard the aircraft platform, to allow for accurate geographic coordinates to be instantly assigned to each photograph. Most camera systems also include mechanisms which compensate for the effect of the aircraft motion relative to the ground, in order to limit distortion as much as possible.

Section 2.7 Cameras and Aerial Photography

When obtaining vertical aerial photographs, the aircraft normally flies in a series of lines, each called a **flight line**. Photos are taken in rapid succession looking straight down at the ground, often with a 50-60 percent overlap (A) between successive photos. The overlap ensures total coverage along a flight line and also facilitates **stereoscopic viewing**. Successive photo pairs display the overlap region from different perspectives and can be viewed through a device called a **stereoscope** to see a three-dimensional view of the area, called a **stereo model**. Many applications of aerial photography use stereoscopic coverage and stereo viewing.



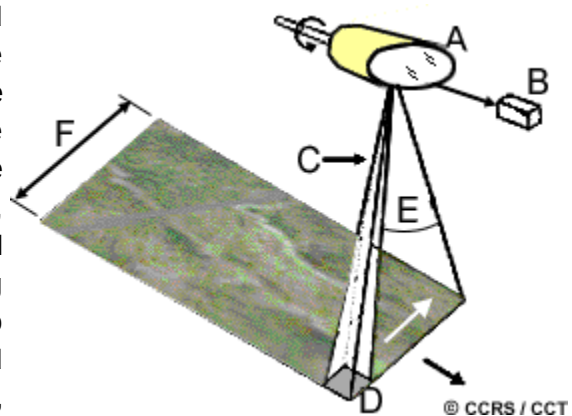
Aerial photographs are most useful when fine spatial detail is more critical than spectral information, as their spectral resolution is generally coarse when compared to data captured with electronic sensing devices. The geometry of vertical photographs is well understood and it is possible to make very accurate measurements from them, for a variety of different applications (geology, forestry, mapping, etc.). The science of making measurements from photographs is called **photogrammetry** and has been performed extensively since the very beginnings of aerial photography. Photos are most often interpreted manually by a human analyst (often viewed stereoscopically). They can also be scanned to create a digital image and then analyzed in a digital computer environment. In Chapter 4, we will discuss in greater detail, various methods (manually and by computer) for interpreting different types of remote sensing images.

Multiband photography uses multi-lens systems with different film-filter combinations to acquire photos simultaneously in a number of different spectral ranges. The advantage of these types of cameras is their ability to record reflected energy separately in discrete wavelength ranges, thus providing potentially better separation and identification of various features. However, simultaneous analysis of these multiple photographs can be problematic. **Digital cameras**, which record electromagnetic radiation electronically, differ significantly from their counterparts which use film. Instead of using film, digital cameras use a gridded array of silicon coated CCDs (charge-coupled devices) that individually respond to electromagnetic radiation. Energy reaching the surface of the CCDs causes the generation of an electronic charge which is proportional in magnitude to the "brightness" of the ground area. A digital number for each spectral band is assigned to each pixel based on the magnitude of the electronic charge. The digital format of the output image is amenable to digital analysis and archiving in a computer environment, as well as output as a hardcopy product similar to regular photos. Digital cameras also provide quicker turn-around for acquisition and retrieval of data and allow greater control of the spectral resolution. Although parameters vary, digital imaging systems are capable of collecting data with a spatial resolution of 0.3m, and with a spectral resolution of 0.012 mm to 0.3 mm. The size of the pixel arrays varies between systems, but typically ranges between 512 x 512 to 2048 x 2048.

2.8 Multispectral Scanning

Many electronic (as opposed to photographic) remote sensors acquire data using **scanning systems**, which employ a sensor with a narrow field of view (i.e. IFOV) that sweeps over the terrain to build up and produce a two-dimensional image of the surface. Scanning systems can be used on both aircraft and satellite platforms and have essentially the same operating principles. A scanning system used to collect data over a variety of different wavelength ranges is called a **multispectral scanner (MSS)**, and is the most commonly used scanning system. There are two main modes or methods of scanning employed to acquire multispectral image data - **across-track scanning**, and **along-track scanning**.

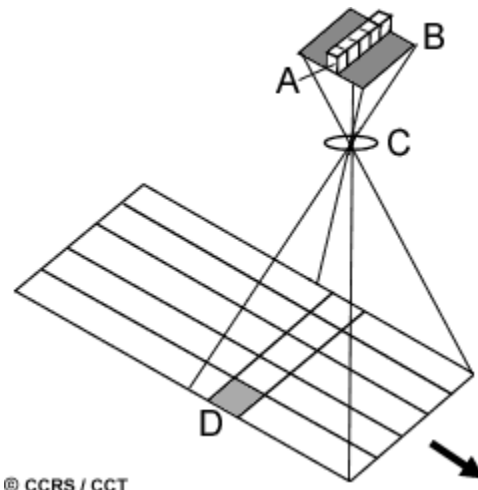
Across-track scanners scan the Earth in a series of lines. The lines are oriented perpendicular to the direction of motion of the sensor platform (i.e. across the swath). Each line is scanned from one side of the sensor to the other, using a **rotating mirror (A)**. As the platform moves forward over the Earth, successive scans build up a two-dimensional image of the Earth's surface. The incoming reflected or emitted radiation is separated into several spectral components that are detected independently. The UV, visible, near-infrared, and thermal radiation are dispersed into their



constituent wavelengths. A bank of internal **detectors (B)**, each sensitive to a specific range of wavelengths, detects and measures the energy for each spectral band and then, as an electrical signal, they are converted to digital data and recorded for subsequent computer processing.

The **IFOV (C)** of the sensor and the altitude of the platform determine the **ground resolution cell viewed (D)**, and thus the spatial resolution. The **angular field of view (E)** is the sweep of the mirror, measured in degrees, used to record a scan line, and determines the width of the imaged **swath (F)**. Airborne scanners typically sweep large angles (between 90° and 120°), while satellites, because of their higher altitude need only to sweep fairly small angles (10-20°) to cover a broad region. Because the distance from the sensor to the target increases towards the edges of the swath, the ground resolution cells also become larger and introduce geometric distortions to the images. Also, the length of time the IFOV "sees" a ground resolution cell as the rotating mirror scans (called the **dwell time**), is generally quite short and influences the design of the spatial, spectral, and radiometric resolution of the sensor.

Along-track scanners also use the forward motion of the platform to record successive scan lines and build up a two-dimensional image, perpendicular to the flight direction. However, instead of a scanning mirror, they use a linear array of detectors (A) located at the focal plane of the image (B) formed by lens systems (C), which are "pushed" along in the flight track direction (i.e. along track). These systems are also referred to as **pushbroom scanners**, as the motion of the detector array is analogous to the bristles of a broom being pushed along a floor. Each individual detector measures the energy for a single ground resolution cell (D) and thus the size and IFOV of the detectors determines



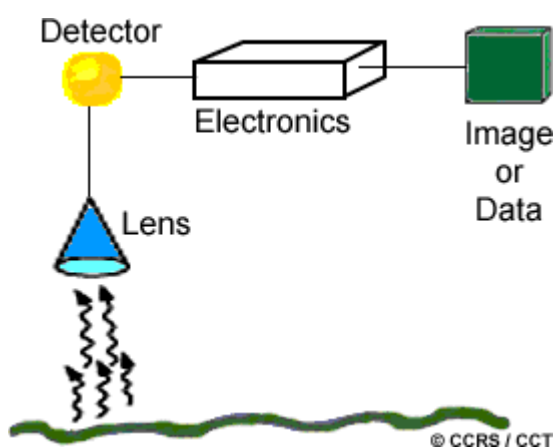
the spatial resolution of the system. A separate linear array is required to measure each spectral band or channel. For each scan line, the energy detected by each detector of each linear array is sampled electronically and digitally recorded.

Along-track scanners with linear arrays have several advantages over across-track mirror scanners. The array of detectors combined with the pushbroom motion allows each detector to "see" and measure the energy from each ground resolution cell for a longer period of time (dwell time). This allows more energy to be detected and improves the radiometric resolution. The increased dwell time also facilitates smaller IFOVs and narrower bandwidths for each detector. Thus, finer spatial and spectral resolution can be achieved without impacting radiometric resolution. Because detectors are usually solid-state microelectronic devices, they are generally smaller, lighter, require less power, and are more reliable and last longer because they have no moving parts. On the other hand, cross-calibrating thousands of detectors to achieve uniform sensitivity across the array is necessary and complicated.

Regardless of whether the scanning system used is either of these two types, it has several advantages over photographic systems. The spectral range of photographic systems is restricted to the visible and near-infrared regions while MSS systems can extend this range into the thermal infrared. They are also capable of much higher spectral resolution than photographic systems. Multi-band or multispectral photographic systems use separate lens systems to acquire each spectral band. This may cause problems in ensuring that the different bands are comparable both spatially and radiometrically and with registration of the multiple images. MSS systems acquire all spectral bands simultaneously through the same optical system to alleviate these problems. Photographic systems record the energy detected by means of a photochemical process which is difficult to measure and to make consistent. Because MSS data are recorded electronically, it is easier to determine the specific amount of energy measured, and they can record over a greater range of values in a digital format. Photographic systems require a continuous supply of film and processing on the ground after the photos have been taken. The digital recording in MSS systems facilitates transmission of data to receiving stations on the ground and immediate processing of data in a computer environment.

2.9 Thermal Imaging

Many multispectral (MSS) systems sense radiation in the thermal infrared as well as the visible and reflected infrared portions of the spectrum. However, remote sensing of energy emitted from the Earth's surface in the thermal infrared ($3\text{ }\mu\text{m}$ to $15\text{ }\mu\text{m}$) is different than the sensing of reflected energy. **Thermal sensors** use photo detectors sensitive to the direct contact of photons on their surface, to detect emitted thermal radiation. The detectors are cooled to temperatures close to absolute zero in order to limit their own thermal emissions. Thermal sensors essentially measure the surface temperature and thermal properties of targets.



Thermal imagers are typically across-track scanners (like those described in the previous section) that detect emitted radiation in only the thermal portion of the spectrum. Thermal sensors employ one or more internal temperature references for comparison with the detected radiation, so they can be related to absolute radiant temperature. The data are generally recorded on film and/or magnetic tape and the temperature resolution of current sensors can reach $0.1\text{ }^{\circ}\text{C}$. For analysis, an image of relative radiant temperatures (a **thermogram**) is depicted in grey levels, with warmer temperatures shown in light tones, and cooler temperatures in dark tones. Imagery which portrays relative temperature differences in their relative spatial locations are sufficient for most applications. Absolute temperature measurements may be calculated but require accurate calibration and measurement of the temperature references and detailed knowledge of the thermal properties of the target, geometric distortions, and radiometric effects.

Because of the relatively long wavelength of thermal radiation (compared to visible radiation), atmospheric scattering is minimal. However, absorption by atmospheric gases normally

Section 2.9 Thermal Imaging

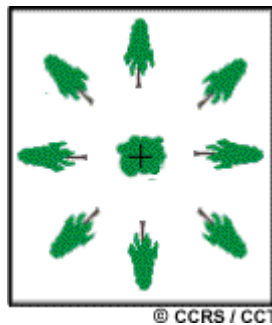
restricts thermal sensing to two specific regions - 3 to 5 μm and 8 to 14 μm . Because energy decreases as the wavelength increases, thermal sensors generally have large IFOVs to ensure that enough energy reaches the detector in order to make a reliable measurement.

Therefore the spatial resolution of thermal sensors is usually fairly coarse, relative to the spatial resolution possible in the visible and reflected infrared. Thermal imagery can be acquired during the day or night (because the radiation is emitted not reflected) and is used for a variety of applications such as military reconnaissance, disaster management (forest fire mapping), and heat loss monitoring.

2.10 Geometric Distortion in Imagery

Any remote sensing image, regardless of whether it is acquired by a multispectral scanner on board a satellite, a photographic system in an aircraft, or any other platform/sensor combination, will have various geometric distortions. This problem is inherent in remote sensing, as we attempt to accurately represent the three-dimensional surface of the Earth as a two-dimensional image. All remote sensing images are subject to some form of geometric distortions, depending on the manner in which the data are acquired. These errors may be due to a variety of factors, including one or more of the following, to name only a few:

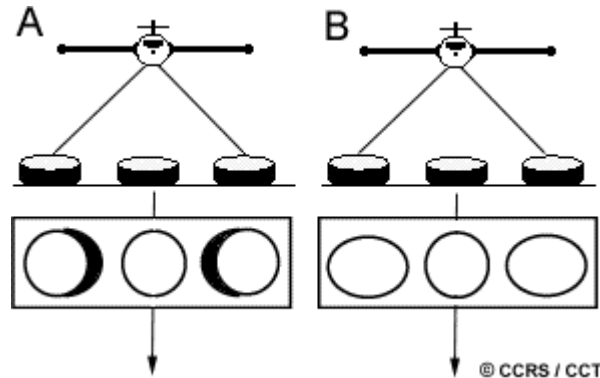
- the perspective of the sensor optics,
- the motion of the scanning system,
- the motion and (in)stability of the platform,
- the platform altitude, attitude, and velocity,
- the terrain relief, and
- the curvature and rotation of the Earth.



Framing systems, such as cameras used for aerial photography, provide an instantaneous "snapshot" view of the Earth from directly overhead. The primary geometric distortion in vertical aerial photographs is due to **relief displacement**. Objects directly below the centre of the camera lens (i.e. at the **nadir**) will have only their tops visible, while all other objects will appear to lean away from the centre of the photo such that their tops and sides are visible. If the objects are tall or are far away from the centre of the photo, the distortion and positional error will be larger.

The geometry of along-track scanner imagery is similar to that of an aerial photograph for each scan line as each detector essentially takes a "snapshot" of each ground resolution cell. Geometric variations between lines are caused by random variations in platform altitude and attitude along the direction of flight.

Section 2.10 Geometric Distortion in Imagery



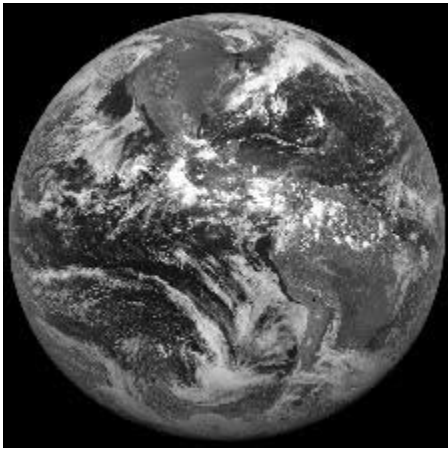
Images from across-track scanning systems exhibit two main types of **geometric distortion**. They too exhibit relief displacement (A), similar to aerial photographs, but in only one direction parallel to the direction of scan. There is no displacement directly below the sensor, at nadir. As the sensor scans across the swath, the top and side of objects are imaged and appear to lean away from the nadir point in each scan line. Again, the displacement increases, moving towards the edges of the swath. Another distortion (B) occurs due to the rotation of the scanning optics. As the sensor scans across each line, the distance from the sensor to the ground increases further away from the centre of the swath. Although the scanning mirror rotates at a constant speed, the IFOV of the sensor moves faster (relative to the ground) and scans a larger area as it moves closer to the edges. This effect results in the compression of image features at points away from the nadir and is called **tangential scale distortion**.

All images are susceptible to geometric distortions caused by variations in platform stability including changes in their speed, altitude, and attitude (angular orientation with respect to the ground) during data acquisition. These effects are most pronounced when using aircraft platforms and are alleviated to a large degree with the use of satellite platforms, as their orbits are relatively stable, particularly in relation to their distance from the Earth. However, the eastward rotation of the Earth, during a satellite orbit causes the sweep of scanning systems to cover an area slightly to the west of each previous scan. The resultant imagery is thus skewed across the image. This is known as **skew distortion** and is common in imagery obtained from satellite multispectral scanners.

The sources of geometric distortion and positional error vary with each specific situation, but are inherent in remote sensing imagery. In most instances, we may be able to remove, or at least reduce these errors but they must be taken into account in each instance before attempting to make measurements or extract further information.

Now that we have learned about some of the general characteristics of platforms and sensors, in the next sections we will look at some specific sensors (primarily satellite systems) operating in the visible and infrared portions of the spectrum.

2.11 Weather Satellites/Sensors



Weather monitoring and forecasting was one of the first civilian (as opposed to military) applications of satellite remote sensing, dating back to the first true weather satellite, TIROS-1 (Television and Infrared Observation Satellite - 1), launched in 1960 by the United States.

Several other weather satellites were launched over the next five years, in near-polar orbits, providing repetitive coverage of global weather patterns. In 1966, NASA (the U.S. National Aeronautics and Space Administration) launched the geostationary Applications Technology Satellite (ATS-1) which provided **hemispheric images** of the Earth's surface and cloud cover every half hour.

For the first time, the development and movement of weather systems could be routinely monitored. Today, several countries operate weather, or meteorological satellites to monitor weather conditions around the globe. Generally speaking, these satellites use sensors which have fairly coarse spatial resolution (when compared to systems for observing land) and provide large areal coverage.

Their temporal resolutions are generally quite high, providing frequent observations of the Earth's surface, atmospheric moisture, and cloud cover, which allows for near-continuous monitoring of global weather conditions, and hence - forecasting. Here we review a few of the representative satellites/sensors used for meteorological applications.

GOES



The GOES (Geostationary Operational Environmental Satellite) System is the follow-up to the ATS series. They were designed by NASA for the National Oceanic and Atmospheric Administration (NOAA) to provide the United States National Weather Service with frequent, small-scale imaging of the Earth's surface and cloud cover. The GOES series of satellites have been used extensively by meteorologists for weather monitoring and forecasting for over 20 years. These satellites are part of a global network of meteorological

satellites spaced at approximately 70° longitude intervals around the Earth in order to provide near-global coverage. Two GOES satellites, placed in **geostationary orbits** 36000 km above the equator, each view approximately one-third of the Earth. One is situated at 75°W longitude and monitors North and South America and most of the Atlantic Ocean. The other is situated at 135°W longitude and monitors North America and the Pacific Ocean basin. Together they

Section 2.11 Weather Satellites/Sensors

cover from 20°W to 165°E longitude. This GOES image covers a portion of the southeastern United States, and the adjacent ocean areas where many severe storms originate and develop. This image shows Hurricane Fran approaching the southeastern United States and the Bahamas in September of 1996.

Two generations of GOES satellites have been launched, each measuring emitted and reflected radiation from which atmospheric temperature, winds, moisture, and cloud cover can be derived. The first generation of satellites consisted of GOES-1 (launched 1975) through GOES-7 (launched 1992). Due to their design, these satellites were capable of viewing the Earth only a small percentage of the time (approximately five per cent). The second generation of satellites began with GOES-8 (launched 1994) and has numerous technological improvements over the first series. They provide near-continuous observation of the Earth allowing more frequent imaging (as often as every 15 minutes). This increase in temporal resolution coupled with improvements in the spatial and radiometric resolution of the sensors provides timelier information and improved data quality for forecasting meteorological conditions.

GOES-8 and the other second generation GOES satellites have separate **imaging** and **sounding** instruments. The **imager** has five channels sensing visible and infrared reflected and emitted solar radiation. The infrared capability allows for day and night imaging. Sensor pointing and scan selection capability enable imaging of an entire hemisphere, or small-scale imaging of selected areas. The latter allows meteorologists to monitor specific weather trouble spots to assist in improved short-term forecasting. The imager data are 10-bit radiometric resolution, and can be transmitted directly to local user terminals on the Earth's surface. The accompanying table describes the individual bands, their spatial resolution, and their meteorological applications.

GOES Bands

Band	Wavelength Range (>μm)	Spatial Resolution	Application
1	0.52 - 0.72 (visible)	1 km	cloud, pollution, and haze detection; severe storm identification
2	3.78 - 4.03 (shortwave IR)	4 km	identification of fog at night; discriminating water clouds and snow or ice clouds during daytime; detecting fires and volcanoes; night time determination of sea surface temperatures
3	6.47 - 7.02 (upper level water vapour)	4 km	estimating regions of mid-level moisture content and advection; tracking mid-level atmospheric motion
4	10.2 - 11.2 (longwave IR)	4 km	identifying cloud-drift winds, severe storms, and heavy rainfall
5	11.5 - 12.5 (IR window sensitive to water vapour)	4 km	identification of low-level moisture; determination of sea surface temperature; detection of airborne dust and volcanic ash

The 19 channel **sounder** measures emitted radiation in 18 thermal infrared bands and reflected radiation in one visible band. These data have a spatial resolution of 8 km and 13-bit radiometric resolution. Sounder data are used for surface and cloud-top temperatures, multi-level moisture profiling in the atmosphere, and ozone distribution analysis.

NOAA AVHRR

NOAA is also responsible for another series of satellites which are useful for meteorological, as well as other, applications. These satellites, in **sun-synchronous, near-polar orbits** (830-870 km above the Earth), are part of the Advanced TIROS series (originally dating back to 1960) and provide complementary information to the geostationary meteorological satellites (such as GOES). Two satellites, each providing global coverage, work together to ensure that data for any region of the Earth is no more than six hours old. One satellite crosses the equator in the early morning from north-to-south while the other crosses in the afternoon.

The primary sensor on board the NOAA satellites, used for both meteorology and small-scale Earth observation and reconnaissance, is the **Advanced Very High Resolution Radiometer (AVHRR)**. The AVHRR sensor detects radiation in the visible, near and mid infrared, and thermal infrared portions of the electromagnetic spectrum, over a swath width of 3000 km. The accompanying table, outlines the AVHRR bands, their wavelengths and spatial resolution (at swath nadir), and general applications of each.

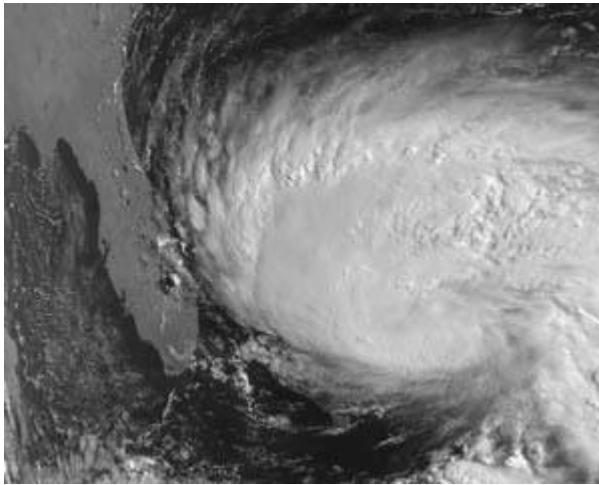
NOAA AVHRR Bands

Band	Wavelength Range (μm)	Spatial Resolution	Application
1	0.58 - 0.68 (red)	1.1 km	cloud, snow, and ice monitoring
2	0.725 - 1.1 (near IR)	1.1 km	water, vegetation, and agriculture surveys
3	3.55 -3.93 (mid IR)	1.1 km	sea surface temperature, volcanoes, and forest fire activity
4	10.3 - 11.3 (thermal IR)	1.1 km	sea surface temperature, soil moisture
5	11.5 - 12.5 (thermal IR)	1.1 km	sea surface temperature, soil moisture

AVHRR data can be acquired and formatted in four operational modes, differing in resolution and method of transmission. Data can be transmitted directly to the ground and viewed as data are collected, or recorded on board the satellite for later transmission and processing. The accompanying table describes the various data formats and their characteristics.

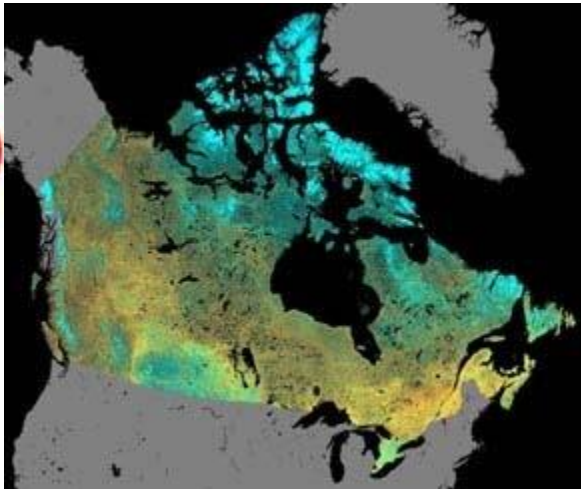
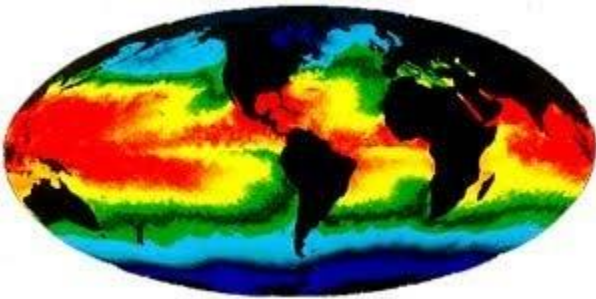
AVHRR Data Formats

Format	Spatial Resolution	Transmission and Processing
APT (Automatic Picture Transmission)	4 km	low-resolution direct transmission and display
HRPT (High Resolution Picture Transmission)	1.1 km	full-resolution direct transmission and display
GAC (Global Area Coverage)	4 km	low-resolution coverage from recorded data
LAC (Local Area Coverage)	1.1 km	selected full-resolution local area data from recorded data



Although AVHRR data are widely used for **weather system** forecasting and analysis, the sensor is also well-suited to observation and monitoring of land features. AVHRR has much coarser spatial resolution than other typical land observations sensors (discussed in the next section), but is used extensively for monitoring regional, small-scale phenomena, including mapping of **sea surface temperature**, and natural vegetation and crop conditions. **Mosaics** covering large areas can be created from several AVHRR data sets allowing small scale analysis and mapping of broad vegetation cover. In Canada, AVHRR data received at the Prince

Albert Receiving Station Saskatchewan, are used as part of a crop information system, monitoring the health of grain crops in the Prairies throughout the growing season.



Other Weather Satellites

The United States operates the **DMSP** (Defense Meteorological Satellite Program) series of satellites which are also used for weather monitoring. These are near-polar orbiting satellites whose Operational Linescan System (OLS) sensor provides twice daily coverage with a swath width of 3000 km at a spatial resolution of 2.7 km. It has two fairly broad wavelength bands: a visible and near infrared band (0.4 to 1.1 μm) and a thermal infrared band (10.0 to 13.4 μm). An interesting feature of the sensor is its ability to acquire visible band night time imagery under very low illumination conditions. With this sensor, it is possible to collect striking images of the Earth showing (typically) the night time lights of large urban centres.

Section 2.11 Weather Satellites/Sensors

There are several other meteorological satellites in orbit, launched and operated by other countries, or groups of countries. These include Japan, with the **GMS** satellite series, and the consortium of European communities, with the **Meteosat** satellites. Both are geostationary satellites situated above the equator over Japan and Europe, respectively. Both provide half-hourly imaging of the Earth similar to GOES. **GMS** has two bands: 0.5 to 0.75 μm (1.25 km resolution), and 10.5 to 12.5 μm (5 km resolution). **Meteosat** has three bands: visible band (0.4 to 1.1 μm ; 2.5 km resolution), mid-IR (5.7 to 7.1 μm ; 5 km resolution), and thermal IR (10.5 to 12.5 μm ; 5 km resolution).

2.12 Land Observation Satellites/Sensors

Landsat



Although many of the weather satellite systems (such as those described in the previous section) are also used for monitoring the Earth's surface, they are not optimized for detailed mapping of the land surface. Driven by the exciting views from, and great success of the early meteorological satellites in the 1960's, as well as from images taken during manned spacecraft missions, the first satellite designed specifically to monitor the Earth's surface, Landsat-1, was launched by NASA in 1972. Initially referred to as ERTS-1, (Earth Resources Technology Satellite), **Landsat** was designed as an

experiment to test the feasibility of collecting multi-spectral Earth observation data from an unmanned satellite platform. Since that time, this highly successful program has collected an abundance of data from around the world from several Landsat satellites. Originally managed by NASA, responsibility for the Landsat program was transferred to NOAA in 1983. In 1985, the program became commercialized, providing data to civilian and applications users.

Landsat's success is due to several factors, including: a combination of sensors with spectral bands tailored to Earth observation; functional spatial resolution; and good areal coverage (swath width and revisit period). The long lifespan of the program has provided a voluminous archive of Earth resource data facilitating long term monitoring and historical records and research. All Landsat satellites are placed in near-polar, sun-synchronous orbits. The first three satellites (Landsats 1-3) are at altitudes around 900 km and have revisit periods of 18 days while the later satellites are at around 700 km and have revisit periods of 16 days. All Landsat satellites have equator crossing times in the morning to optimize illumination conditions.

A number of sensors have been on board the Landsat series of satellites, including the **Return Beam Vidicon (RBV)** camera systems, the **MultiSpectral Scanner (MSS)** systems, and the **Thematic Mapper (TM)**. The most popular instrument in the early days of Landsat was the MultiSpectral Scanner (MSS) and later the Thematic Mapper (TM). Each of these sensors collected data over a swath width of 185 km, with a full scene being defined as 185 km x 185 km.

The MSS senses the electromagnetic radiation from the Earth's surface in four spectral bands. Each band has a spatial resolution of approximately 60 x 80 metres and a radiometric resolution of 6 bits, or 64 digital numbers. Sensing is accomplished with a line scanning device using an oscillating mirror. Six scan lines are collected simultaneously with each west-to-east sweep of the scanning mirror. The accompanying table outlines the spectral wavelength ranges for the MSS.

MSS Bands

Channel		Wavelength Range (μm)
Landsat 1,2,3	Landsat 4,5	
MSS 4	MSS 1	0.5 - 0.6 (green)
MSS 5	MSS 2	0.6 - 0.7 (red)
MSS 6	MSS 3	0.7 - 0.8 (near infrared)
MSS 7	MSS 4	0.8 - 1.1 (near infrared)

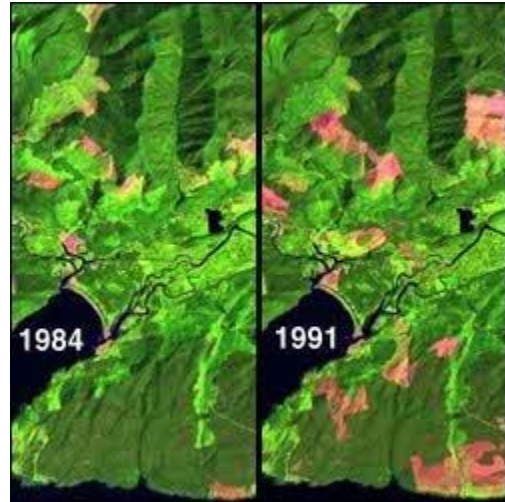
Routine collection of MSS data ceased in 1992, as the use of TM data, starting on Landsat 4, superseded the MSS. The **TM** sensor provides several improvements over the MSS sensor including: higher spatial and radiometric resolution; finer spectral bands; seven as opposed to four spectral bands; and an increase in the number of detectors per band (16 for the non-thermal channels versus six for MSS). Sixteen scan lines are captured simultaneously for each non-thermal spectral band (four for thermal band), using an oscillating mirror which scans during both the forward (west-to-east) and reverse (east-to-west) sweeps of the scanning mirror. This difference from the MSS increases the **dwell time** (see section 2.8) and improves the geometric and radiometric integrity of the data. Spatial resolution of TM is 30 m for all but the thermal infrared band which is 120 m. All channels are recorded over a range of 256 digital numbers (8 bits). The accompanying table outlines the spectral resolution of the individual TM bands and some useful applications of each.

TM Bands

Channel	Wavelength Range (μm)	Application
TM 1	0.45 - 0.52 (blue)	soil/vegetation discrimination; bathymetry/coastal mapping; cultural/urban feature identification
TM 2	0.52 - 0.60 (green)	green vegetation mapping (measures reflectance peak); cultural/urban feature identification
TM 3	0.63 - 0.69 (red)	vegetated vs. non-vegetated and plant species discrimination (plant chlorophyll absorption); cultural/urban feature identification
TM 4	0.76 - 0.90 (near IR)	identification of plant/vegetation types, health, and biomass content; water body delineation; soil moisture
TM 5	1.55 - 1.75 (short wave IR)	sensitive to moisture in soil and vegetation; discriminating snow and cloud-covered areas
TM 6	10.4 - 12.5 (thermal IR)	vegetation stress and soil moisture discrimination related to thermal radiation; thermal mapping (urban, water)
TM 7	2.08 - 2.35 (short wave IR)	discrimination of mineral and rock types; sensitive to vegetation moisture content

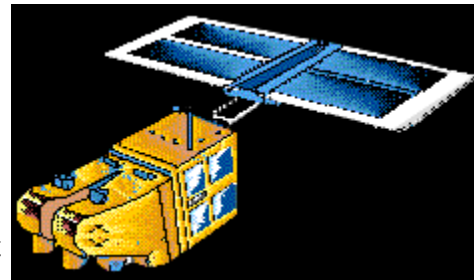
Section 2.12 Land Observation Satellites/Sensors

Data from both the TM and MSS sensors are used for a wide variety of applications, including resource management, mapping, environmental monitoring, and change detection (e.g. **monitoring forest clearcutting**). The archives of Canadian imagery include over 350,000 scenes of MSS and over 200,000 scenes of TM, managed by the licensed distributor in Canada: RSI Inc. Many more scenes are held by foreign facilities around the world.



SPOT

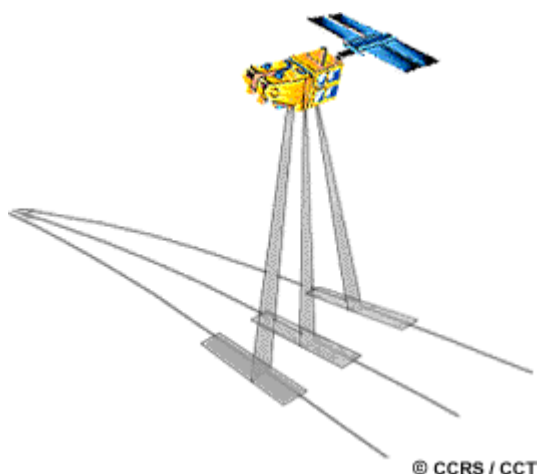
SPOT (Système Pour l'Observation de la Terre) is a series of Earth observation imaging satellites designed and launched by CNES (Centre National d'Études Spatiales) of France, with support from Sweden and Belgium. SPOT-1 was launched in 1986, with successors following every three or four years. All satellites are in sun-synchronous, near-polar orbits at altitudes around 830 km above the Earth, which results in orbit repetition every 26 days. They have equator crossing times around 10:30 AM local solar time. SPOT was designed to be a commercial provider of Earth observation data, and was the first satellite to use along-track, or pushbroom scanning technology.



The SPOT satellites each have twin **high resolution visible (HRV)** imaging systems, which can be operated independently and simultaneously. Each HRV is capable of sensing either in a high spatial resolution single-channel **panchromatic (PLA)** mode, or a coarser spatial resolution three-channel **multispectral (MLA)** mode. Each along-track scanning HRV sensor consists of four linear arrays of detectors: one 6000 element array for the panchromatic mode recording at a spatial resolution of 10 m, and one 3000 element array for each of the three multispectral bands, recording at 20 m spatial resolution. The swath width for both modes is 60 km at nadir. The accompanying table illustrates the spectral characteristics of the two different modes.

HRV Mode Spectral Ranges

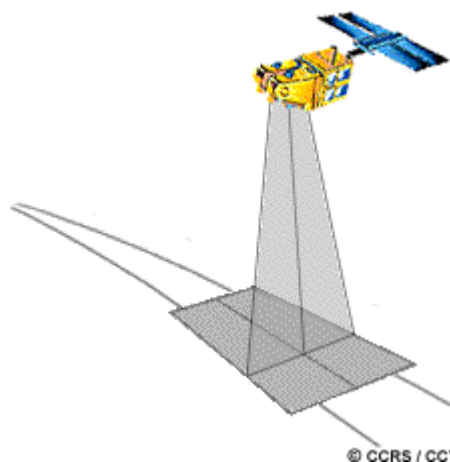
Mode/Band	Wavelength Range (μm)
Panchromatic (PLA)	0.51 - 0.73 (blue-green-red)
Multispectral (MLA)	
Band 1	0.50 - 0.59 (green)
Band 2	0.61 - 0.68 (red)
Band 3	0.79 - 0.89 (near infrared)



The viewing angle of the sensors can be adjusted to look to either side of the satellite's vertical (nadir) track, allowing **off-nadir viewing** which increases the satellite's revisit capability. This ability to point the sensors up to 27° from nadir, allows SPOT to view within a 950 km swath and to revisit any location several times per week. As the sensors point away from nadir, the swath varies from 60 to 80 km in width. This not only improves the ability to monitor specific locations and increases the chances of obtaining cloud free scenes, but the off-nadir viewing also provides the

capability of acquiring imagery for stereoscopic coverage. By recording the same area from two different angles, the imagery can be viewed and analyzed as a three dimensional model, a technique of tremendous value for terrain interpretation, mapping, and visual terrain simulations.

This oblique viewing capability increases the revisit frequency of equatorial regions to three days (seven times during the 26 day orbital cycle). Areas at a latitude of 45° can be imaged more frequently (11 times in 26 days) due to the convergence of orbit paths towards the poles. By pointing both HRV sensors to cover **adjacent ground swaths** at nadir, a swath of 117 km (3 km overlap between the two swaths) can be imaged. In this mode of operation, either panchromatic or multispectral data can be collected, but not both simultaneously.



Section 2.12 Land Observation Satellites/Sensors

SPOT has a number of benefits over other spaceborne optical sensors. Its fine spatial resolution and pointable sensors are the primary reasons for its popularity. The three-band multispectral data are well suited to displaying as false-colour images and the panchromatic band can also be used to "sharpen" the spatial detail in the multispectral data. SPOT allows applications requiring fine spatial detail (such as **urban mapping**) to be addressed while retaining the cost and timeliness advantage of satellite data. The potential applications of SPOT data are numerous. Applications requiring frequent monitoring (agriculture, forestry) are well served by the SPOT sensors. The acquisition of stereoscopic imagery from SPOT has played an important role in mapping applications and in the derivation of topographic information (Digital Elevation Models - DEMs) from satellite data.



IRS

The Indian Remote Sensing (IRS) satellite series, combines features from both the Landsat MSS/TM sensors and the SPOT HRV sensor. The third satellite in the series, IRS-1C, launched in December, 1995 has three sensors: a single-channel panchromatic (PAN) high resolution camera, a medium resolution four-channel Linear Imaging Self-scanning Sensor (LISS-III), and a coarse resolution two-channel Wide Field Sensor (WiFS). The accompanying table outlines the specific characteristics of each sensor.

IRS Sensors

Sensor	Wavelength Range (μm)	Spatial Resolution	Swath Width	Revisit Period (at equator)
PAN	0.5 - 0.75	5.8 m	70 km	24 days
LISS-II				
Green	0.52 - 0.59	23 m	142 km	24 days
Red	0.62 - 0.68	23 m	142 km	24 days
Near IR	0.77 - 0.86	23 m	142 km	24 days
Shortwave IR	1.55 - 1.70	70 m	148 km	24 days
WiFS				
Red	0.62 - 0.68	188 m	774 km	5 days
Near IR	0.77 - 0.86	188 m	774 km	5 days

In addition to its high spatial resolution, the panchromatic sensor can be steered up to 26° across-track, enabling stereoscopic imaging and increased revisit capabilities (as few as five days), similar to SPOT. This high resolution data is useful for urban planning and mapping applications. The four LISS-III multispectral bands are similar to Landsat's TM bands 1 to 4 and are excellent for vegetation discrimination, land-cover mapping, and natural resource planning. The WiFS sensor is similar to NOAA AVHRR bands and the spatial resolution and coverage is useful for regional scale vegetation monitoring.

MEIS-II and CASI

Although this tutorial concentrates on satellite-borne sensors, it is worth mentioning a couple of Canadian airborne sensors which have been used for various remote sensing applications, as these systems (and others like them) have influenced the design and development of satellite systems. The first is the MEIS-II (**Multispectral Electro-optical Imaging Scanner**) sensor developed for the Canada Centre for Remote Sensing. Although no longer active, MEIS was the first operational use of pushbroom, or along-track scanning technology in an airborne platform. The sensor collected 8-bit data (256 digital numbers) in eight spectral bands ranging from 0.39 to 1.1 μm , using linear arrays of 1728 detectors per band. The specific wavelength ranges were selectable, allowing different band combinations to be used for different applications. Stereo imaging from a single flight line was also possible, with channels aimed ahead of and behind nadir, supplementing the other nadir facing sensors. Both the stereo mapping and the selectable band capabilities were useful in research and development which was applied to development of other satellite (and airborne) sensor systems.

CASI, the **Compact Airborne Spectrographic Imager**, is a leader in airborne imaging, being the first commercial imaging spectrometer. This hyperspectral sensor detects a vast array of narrow spectral bands in the visible and infrared wavelengths, using along-track scanning.

The spectral range covered by the 288 channels is between 0.4 and 0.9 μm . Each band covers a wavelength range of 0.018 μm . While spatial resolution depends on the altitude of the aircraft, the spectral bands measured and the bandwidths used are all programmable to meet the user's specifications and requirements. Hyperspectral sensors such as this can be important sources of diagnostic information about specific targets' absorption and reflection characteristics, in effect providing a spectral 'fingerprint'. Experimentation with CASI and other airborne imaging spectrometers has helped guide the development of hyperspectral sensor systems for advanced satellite systems.

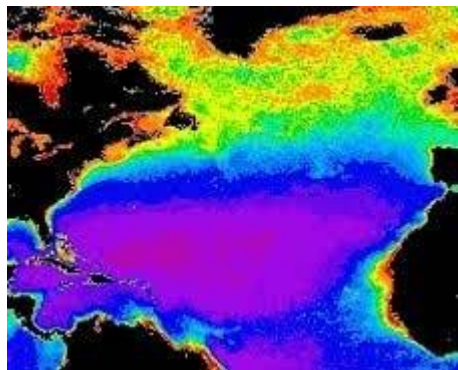
2.13 Marine Observation Satellites/Sensors

The Earth's oceans cover more than two-thirds of the Earth's surface and play an important role in the global climate system. They also contain an abundance of living organisms and natural resources which are susceptible to pollution and other man-induced hazards. The meteorological and land observations satellites/sensors we discussed in the previous two sections can be used for monitoring the oceans of the planet, but there are other satellite/sensor systems which have been designed specifically for this purpose.

The Nimbus-7 satellite, launched in 1978, carried the first sensor, the **Coastal Zone Colour Scanner (CZCS)**, specifically intended for monitoring the Earth's oceans and water bodies. The primary objective of this sensor was to observe ocean colour and temperature, particularly in coastal zones, with sufficient spatial and spectral resolution to detect pollutants in the upper levels of the ocean and to determine the nature of materials suspended in the water column. The Nimbus satellite was placed in a sun-synchronous, near-polar orbit at an altitude of 955 km. Equator crossing times were local noon for ascending passes and local midnight for descending passes. The repeat cycle of the satellite allowed for global coverage every six days, or every 83 orbits. The CZCS sensor consisted of six spectral bands in the visible, near-IR, and thermal portions of the spectrum each collecting data at a spatial resolution of 825 m at nadir over a 1566 km swath width. The accompanying table outlines the spectral ranges of each band and the primary parameter measured by each.

CZCS Spectral Bands

Channel	Wavelength Range (μm)	Primary Measured Parameter
1	0.43 - 0.45	Chlorophyll absorption
2	0.51 - 0.53	Chlorophyll absorption
3	0.54 - 0.56	Gelbstoffe (yellow substance)
4	0.66 - 0.68	Chlorophyll concentration
5	0.70 - 0.80	Surface vegetation
6	10.5 - 12.50	Surface temperature



As can be seen from the table, the first four bands of the CZCS sensor are very narrow. They were optimized to allow detailed discrimination of differences in water reflectance due to **phytoplankton concentrations** and other suspended particulates in the water. In addition to detecting surface vegetation on the water, band 5 was used to discriminate water from land prior to processing the other bands of information. The CZCS sensor ceased operation in 1986.

MOS

The first Marine Observation Satellite (MOS-1) was launched by Japan in February, 1987 and was followed by its successor, MOS-1b, in February of 1990. These satellites carry three different sensors: a four-channel Multispectral Electronic Self-Scanning Radiometer (MESSR), a four-channel Visible and Thermal Infrared Radiometer (VTIR), and a two-channel Microwave Scanning Radiometer (MSR), in the microwave portion of the spectrum. The characteristics of the two sensors in the visible/infrared are described in the accompanying table.

MOS Visible/Infrared Instruments

Sensor	Wavelength Ranges (μm)	Spatial Resolution	Swath Width
MESSR	0.51 - 0.59	50 m	100 km
	0.61 - 0.69	50 m	100 km
	0.72 - 0.80	50 m	100 km
	0.80 - 1.10	50 m	100 km
VTIR	0.50 - 0.70	900 m	1500 km
	6.0 - 7.0	2700 m	1500 km
	10.5 - 11.5	2700 m	1500 km
	11.5 - 12.5	2700 m	1500 km

The MESSR bands are quite similar in spectral range to the Landsat MSS sensor and are thus useful for land applications in addition to observations of marine environments. The MOS systems orbit at altitudes around 900 km and have revisit periods of 17 days.

SeaWiFS

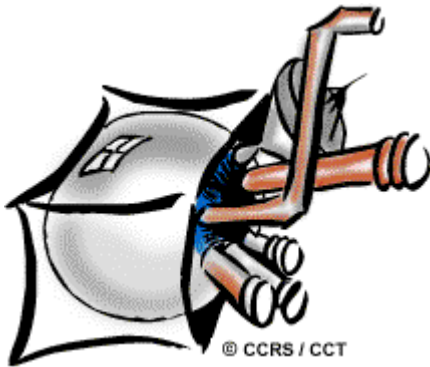
The SeaWiFS (Sea-viewing Wide-Field-of View Sensor) on board the SeaStar spacecraft is an advanced sensor designed for ocean monitoring. It consists of eight spectral bands of very narrow wavelength ranges (see accompanying table) tailored for very specific detection and monitoring of various ocean phenomena including: ocean primary production and phytoplankton processes, ocean influences on climate processes (heat storage and aerosol formation), and monitoring of the cycles of carbon, sulfur, and nitrogen. The orbit altitude is 705 km with a local equatorial crossing time of 12 PM. Two combinations of spatial resolution and swath width are available for each band: a higher resolution mode of 1.1 km (at nadir) over a swath of 2800 km, and a lower resolution mode of 4.5 km (at nadir) over a swath of 1500 km.

SeaWiFS Spectral Bands

Channel	Wavelength Ranges (μm)
1	0.402 - 0.422
2	0.433 - 0.453
3	0.480 - 0.500
4	0.500 - 0.520
5	0.545 - 0.565
6	0.660 - 0.680
7	0.745 - 0.785
8	0.845 - 0.885

These ocean-observing satellite systems are important for global and regional scale monitoring of ocean pollution and health, and assist scientists in understanding the influence and impact of the oceans on the global climate system.

2.14 Other Sensors



The three previous sections provide a representative overview of specific systems available for remote sensing in the (predominantly) optical portions of the electromagnetic spectrum. However, there are many **other types of less common sensors** which are used for remote sensing purposes. We briefly touch on a few of these other types of sensors. The information is not considered comprehensive but serves as an introduction to alternative imagery sources and imaging concepts.

Video

Although coarser in spatial resolution than traditional photography or digital imaging, video cameras provide a useful means of acquiring timely and inexpensive data and vocally annotated imagery. Applications with these requirements include natural disaster management, (fires, flooding), crop and disease assessment, environmental hazard control, and police surveillance. Cameras used for video recording measure radiation in the visible, near infrared, and sometimes mid-infrared portions of the EM spectrum. The image data are recorded onto cassette, and can be viewed immediately.

FLIR

Forward Looking InfraRed (FLIR) systems operate in a similar manner to across-track thermal imaging sensors, but provide an oblique rather than nadir perspective of the Earth's surface. Typically positioned on aircraft or helicopters, and imaging the area ahead of the platform, FLIR systems provide relatively high spatial resolution imaging that can be used for military applications, search and rescue operations, law enforcement, and forest fire monitoring.

Laser fluorosensor

Some targets fluoresce, or emit energy, upon receiving incident energy. This is not a simple reflection of the incident radiation, but rather an absorption of the initial energy, excitation of the molecular components of the target materials, and emission of longer wavelength radiation which is then measured by the sensor. Laser fluorosensors illuminate the target with a specific wavelength of radiation and are capable of detecting multiple wavelengths of fluoresced radiation. This technology has been proven for ocean applications, such as chlorophyll mapping, and pollutant detection, particularly for naturally occurring and accidental oil slicks.

Lidar

Lidar is an acronym for Light Detection And Ranging, an active imaging technology very similar to RADAR (see next paragraph). Pulses of laser light are emitted from the sensor and

Section 2.14 Other Sensors

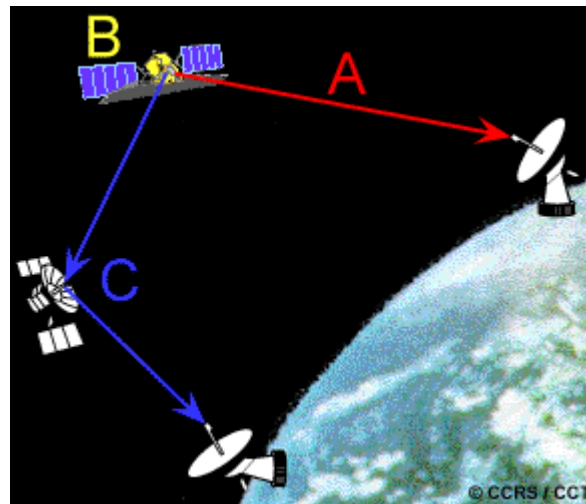
energy reflected from a target is detected. The time required for the energy to reach the target and return to the sensor determines the distance between the two. Lidar is used effectively for measuring heights of features, such as forest canopy height relative to the ground surface, and water depth relative to the water surface (laser profilometer). Lidar is also used in atmospheric studies to examine the particle content of various layers of the Earth's atmosphere and acquire air density readings and monitor air currents.

RADAR

RADAR stands for Radio Detection and Ranging. RADAR systems are active sensors which provide their own source of electromagnetic energy. Active radar sensors, whether airborne or spaceborne, emit microwave radiation in a series of pulses from an antenna, looking obliquely at the surface perpendicular to the direction of motion. When the energy reaches the target, some of the energy is reflected back towards the sensor. This backscattered microwave radiation is detected, measured, and timed. The time required for the energy to travel to the target and return back to the sensor determines the distance or range to the target. By recording the range and magnitude of the energy reflected from all targets as the system passes by, a two-dimensional image of the surface can be produced. Because RADAR provides its own energy source, images can be acquired day or night. Also, microwave energy is able to penetrate through clouds and most rain, making it an all-weather sensor. Because of the unique characteristics and applications of microwave remote sensing, Chapter 3 covers this topic in detail, concentrating on RADAR remote sensing.

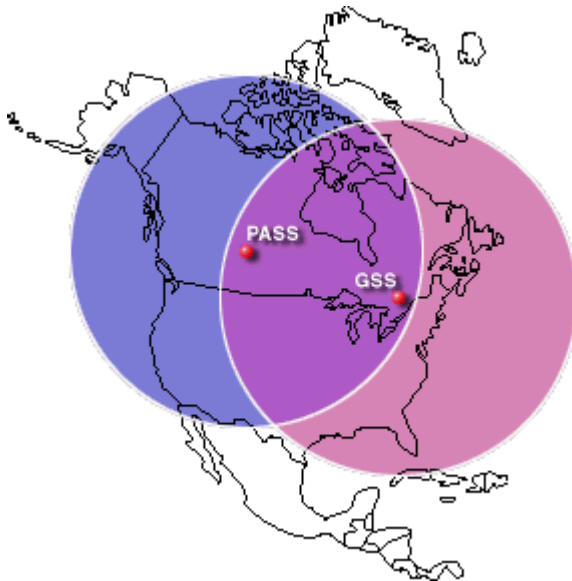
2.15 Data Reception, Transmission, and Processing

Data obtained during airborne remote sensing missions can be retrieved once the aircraft lands. It can then be processed and delivered to the end user. However, data acquired from satellite platforms need to be electronically transmitted to Earth, since the satellite continues to stay in orbit during its operational lifetime. The technologies designed to accomplish this can also be used by an aerial platform if the data are urgently needed on the surface.



There are three main options for **transmitting data** acquired by satellites to the surface. The data can be directly transmitted to Earth if a Ground Receiving Station (GRS) is in the line of sight of the satellite (A). If this is not the case, the data can be recorded on board the satellite (B) for transmission to a GRS at a later time. Data can also be relayed to the GRS through the Tracking and Data Relay Satellite System (TDRSS) (C), which consists of a series of communications satellites in geosynchronous orbit. The data are transmitted from one satellite to another until they reach the appropriate GRS.

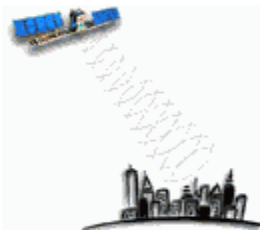
In Canada, CCRS operates two **ground receiving stations** - one at Cantley, Québec (GSS), just outside of Ottawa, and another one at Prince Albert, Saskatchewan (PASS). The combined coverage circles for these Canadian ground stations enable the potential for reception of real-time or recorded data from satellites passing over almost any part of Canada's land mass, and much of the continental United States as well. Other ground stations have been set up around the world to capture data from a variety of satellites.



The data are received at the GRS in a raw digital format. They may then, if required, be processed to correct systematic, geometric and atmospheric distortions to the imagery, and be translated into a standardized format. The data are written to some form of storage medium such as tape, disk or CD. The data are typically archived at most receiving and processing stations, and full libraries of data are managed by government agencies as well as commercial companies responsible for each sensor's archives.

For many sensors it is possible to provide customers with **quick-turnaround** imagery when they need data as quickly as possible after it is collected. Near real-time processing systems are used to produce low resolution imagery in hard copy or soft copy (digital) format within hours of data acquisition. Such imagery can then be faxed or transmitted digitally to end users. One application of this type of fast data processing is to provide imagery to ships sailing in the Arctic, as it allows them to assess current ice conditions quickly in order to make navigation decisions about the easiest/safest routes through the ice. Real-time processing of imagery in airborne systems has been used, for example, to pass thermal infrared imagery to forest fire fighters right at the scene.

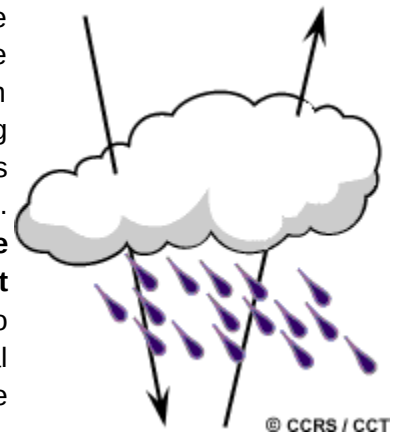
Low resolution quick-look imagery is used to preview archived imagery prior to purchase. The spatial and radiometric quality of these types of data products is degraded, but they are useful for ensuring that the overall quality, coverage and cloud cover of the data is appropriate.



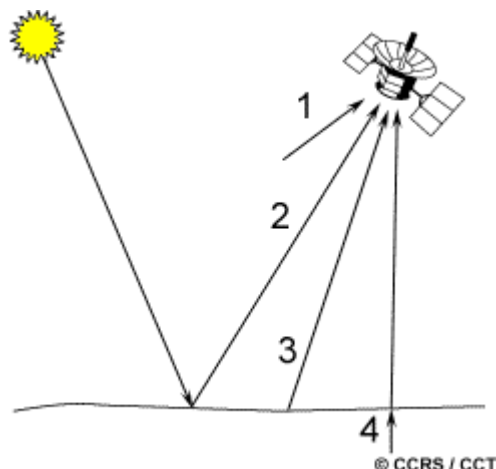
1. Microwave Remote Sensing

3.1 Introduction

Microwave sensing encompasses both active and passive forms of remote sensing. As described in Chapter 2, the microwave portion of the spectrum covers the range from approximately 1cm to 1m in wavelength. Because of their long wavelengths, compared to the visible and infrared, microwaves have special properties that are important for remote sensing. **Longer wavelength microwave radiation can penetrate through cloud cover, haze, dust, and all but the heaviest rainfall** as the longer wavelengths are not susceptible to atmospheric scattering which affects shorter optical wavelengths. This property allows detection of microwave energy under almost all weather and environmental conditions so that data can be collected at any time.



Passive microwave sensing is similar in concept to thermal remote sensing. All objects emit microwave energy of some magnitude, but the amounts are generally very small. A passive microwave sensor detects the naturally emitted microwave energy within its field of view. This emitted energy is related to the temperature and moisture properties of the emitting object or surface. Passive microwave sensors are typically radiometers or scanners and operate in much the same manner as systems discussed previously except that an antenna is used to detect and record the microwave energy.

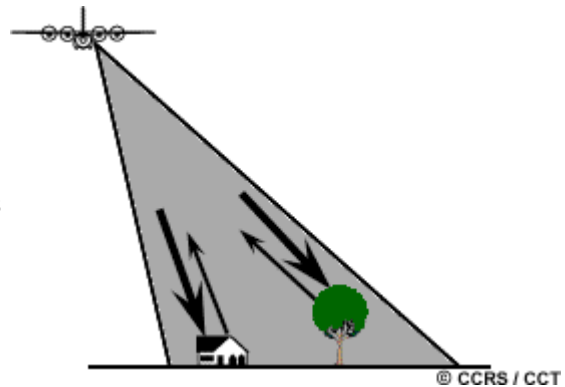


Section 3.1 Microvaves Introduction

The microwave energy recorded by a passive sensor can be emitted by the atmosphere (1), reflected from the surface (2), emitted from the surface (3), or transmitted from the subsurface (4). Because the wavelengths are so long, the energy available is quite small compared to optical wavelengths. Thus, the fields of view must be large to detect enough energy to record a signal. Most passive microwave sensors are therefore characterized by low spatial resolution.

Applications of passive microwave remote sensing include meteorology, hydrology, and oceanography. By looking "at", or "through" the atmosphere, depending on the wavelength, meteorologists can use passive microwaves to measure atmospheric profiles and to determine water and ozone content in the atmosphere. Hydrologists use passive microwaves to measure soil moisture since microwave emission is influenced by moisture content. Oceanographic applications include mapping sea ice, currents, and surface winds as well as detection of pollutants, such as oil slicks.

Active microwave sensors provide their own source of microwave radiation to illuminate the target. Active microwave sensors are generally divided into two distinct categories: **imaging and non-imaging**. The most common form of imaging active microwave sensors is RADAR. **RADAR** is an acronym for **R**adio **D**etection **A**nd **R**anging, which essentially characterizes the function and operation of a radar sensor. The sensor transmits a microwave (radio) signal towards the target and detects the backscattered portion of the signal.



The strength of the backscattered signal is measured to discriminate between different targets and the time delay between the transmitted and reflected signals determines the distance (or **range**) to the target.

Non-imaging microwave sensors include **altimeters** and **scatterometers**. In most cases these are profiling devices which take measurements in one linear dimension, as opposed to the two-dimensional representation of imaging sensors. Radar altimeters transmit short microwave pulses and measure the round trip time delay to targets to determine their distance from the sensor. Generally altimeters look straight down at nadir below the platform and thus measure height or elevation (if the altitude of the platform is accurately known). Radar altimetry is used on aircraft for altitude determination and on aircraft and satellites for topographic mapping and sea surface height estimation. Scatterometers are also generally non-imaging sensors and are used to make precise quantitative measurements of the amount of energy backscattered from targets. The amount of energy backscattered is dependent on the surface properties (roughness) and the angle at which the microwave energy strikes the target. Scatterometry measurements over ocean surfaces can be used to estimate wind speeds based on the sea surface roughness. Ground-based scatterometers are used extensively to accurately measure the backscatter from various targets in order to

Section 3.1 Microvaves Introduction

characterize different materials and surface types. This is analogous to the concept of spectral reflectance curves in the optical spectrum.

For the remainder of this chapter we focus solely on **imaging radars**. As with passive microwave sensing, a major advantage of radar is the capability of the radiation to penetrate through cloud cover and most weather conditions. Because radar is an active sensor, it can also be used to image the surface at any time, day or night. These are the two primary advantages of radar: **all-weather and day or night** imaging. It is also important to understand that, because of the fundamentally different way in which an active radar operates compared to the passive sensors we described in Chapter 2, a radar image is quite different from and has special properties unlike images acquired in the visible and infrared portions of the spectrum. Because of these differences, radar and optical data can be complementary to one another as they offer different perspectives of the Earth's surface providing different information content. We will examine some of these fundamental properties and differences in more detail in the following sections.

Before we delve into the peculiarities of radar, let's first look briefly at the origins and history of imaging radar, with particular emphasis on the Canadian experience in radar remote sensing. The first demonstration of the transmission of radio microwaves and reflection from various objects was achieved by Hertz in 1886. Shortly after the turn of the century, the first rudimentary radar was developed for ship detection. In the 1920s and 1930s, experimental ground-based pulsed radars were developed for detecting objects at a distance. The first imaging radars used during World War II had rotating sweep displays which were used for detection and positioning of aircrafts and ships. After World War II, side-looking airborne radar (SLAR) was developed for military terrain reconnaissance and surveillance where a strip of the ground parallel to and offset to the side of the aircraft was imaged during flight. In the 1950s, advances in SLAR and the development of higher resolution synthetic aperture radar (SAR) were developed for military purposes. In the 1960s these radars were declassified and began to be used for civilian mapping applications. Since this time the development of several airborne and spaceborne radar systems for mapping and monitoring applications use has flourished.

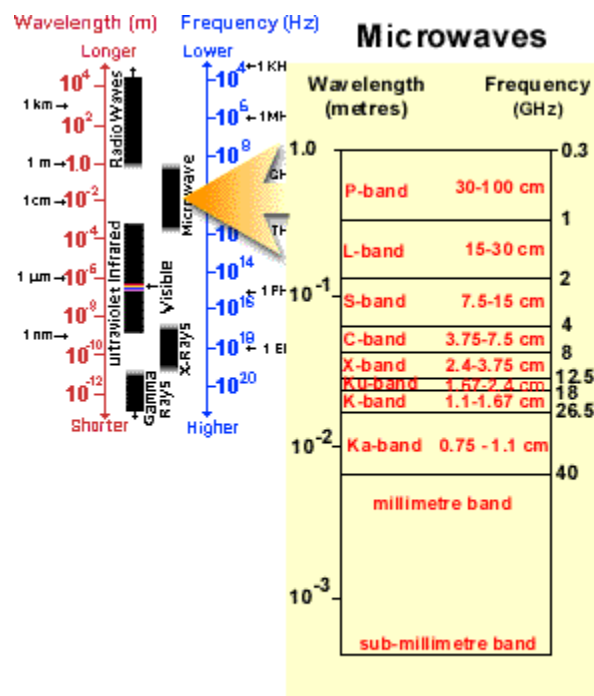
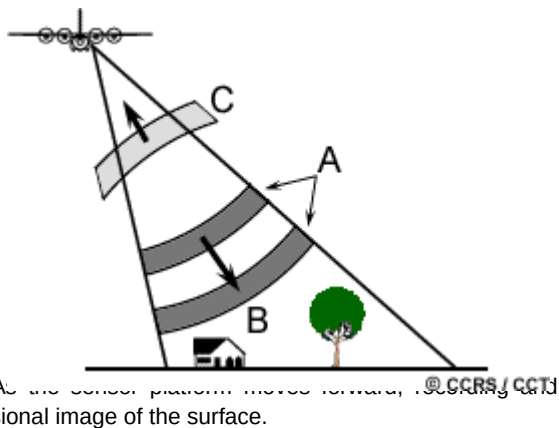
Canada initially became involved in radar remote sensing in the mid-1970s. It was recognized that radar may be particularly well-suited for surveillance of our vast northern expanse, which is often cloud-covered and shrouded in darkness during the Arctic winter, as well as for monitoring and mapping our natural resources. Canada's SURSAT (Surveillance Satellite) project from 1977 to 1979 led to our participation in the (U.S.) SEASAT radar satellite, the first operational civilian radar satellite. The Convair-580 airborne radar program, carried out by the Canada Centre for Remote Sensing following the SURSAT program, in conjunction with radar research programs of other agencies such as NASA and the European Space Agency (ESA), led to the conclusion that spaceborne remote sensing was feasible. In 1987, the Radar Data Development Program (RDDP), was initiated by the Canadian government with the objective of "operationalizing the use of radar data by Canadians". Over the 1980s and early 1990s, several research and commercial airborne radar systems have collected vast amounts of

Section 3.1 Microvaves Introduction

imagery throughout the world demonstrating the utility of radar data for a variety of applications. With the launch of ESA's ERS-1 in 1991, spaceborne radar research intensified, and was followed by the major launches of Japan's J-ERS satellite in 1992, ERS-2 in 1995, and Canada's advanced RADARSAT satellite, also in 1995.

3.2 Radar Basics

As noted in the previous section, a **radar** is essentially a ranging or distance measuring device. It consists fundamentally of a transmitter, a receiver, an antenna, and an electronics system to process and record the data. The transmitter generates successive short bursts (or **pulses** of microwave (A) at regular intervals which are focused by the antenna into a beam (B). The radar beam illuminates the surface obliquely at a right angle to the motion of the platform. The antenna receives a portion of the transmitted energy reflected (or **backscattered**) from various objects within the illuminated beam (C). By measuring the time delay between the transmission of a pulse and the reception of the backscattered "echo" from different targets, their distance from the radar and thus their location can be determined. As the sensor platform moves forward, processing of the backscattered signals builds up a two-dimensional image of the surface.



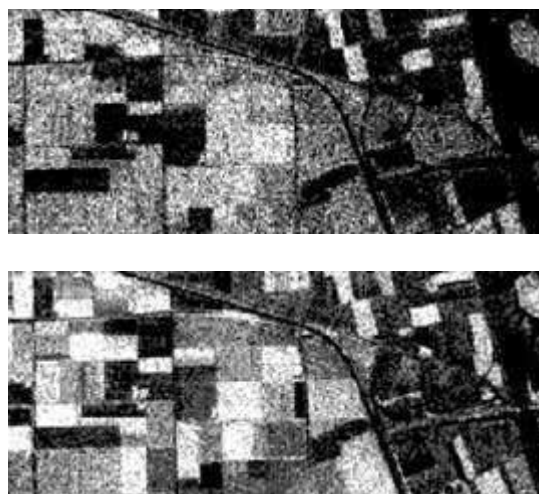
© CCRS / CCT

While we have characterized electromagnetic radiation in the visible and infrared portions of the spectrum primarily by wavelength, microwave portions of the spectrum are often referenced according to both wavelength and frequency. The **microwave region of the spectrum** is quite large, relative to the visible and infrared, and there are several wavelength ranges or bands commonly used which given code letters during World War II, and remain to this day.

- Ka, K, and Ku bands: very short wavelengths used in early airborne radar systems but uncommon today.
- X-band: used extensively on airborne systems for military reconnaissance and terrain mapping.
- C-band: common on many airborne research systems (CCRS Convair-580 and NASA AirSAR) and spaceborne systems (including
- ERS-1 and 2 and RADARSAT).

Section 3.2 Radar Basics

- S-band: used on board the Russian ALMAZ satellite.
- L-band: used onboard American SEASAT and Japanese JERS-1 satellites and NASA airborne system.
- P-band: longest radar wavelengths, used on NASA experimental airborne research system.

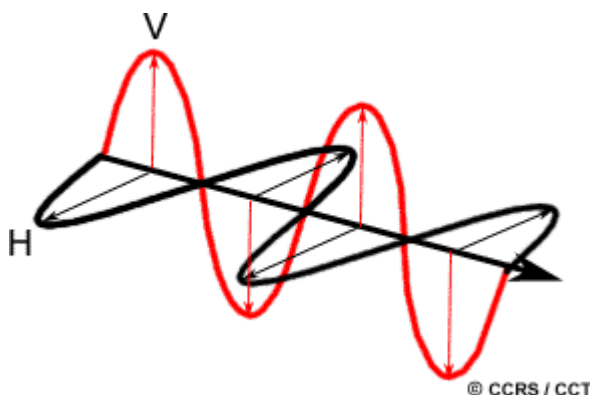


Two radar images of the same agricultural fields

Here are two radar images of the same agricultural fields, each image having been collected using a different radar band. The one on the top was acquired by a C-band radar and the one below was acquired by an L-band radar. You can clearly see that there are significant differences between the way the various fields and crops appear in each of the two images. This is due to the different ways in which the radar energy interacts with the fields and crops depending on the radar wavelength. We will learn more about this in later sections.

When discussing microwave energy, the **polarization** of the radiation is also important. Polarization refers to the orientation of the electric field (recall the definition of electromagnetic radiation from Chapter 1). Most radars are designed to transmit microwave radiation either horizontally polarized (H) or vertically polarized (V). Similarly, the antenna receives either the horizontally or vertically polarized backscattered energy, and some radars can receive both. These two polarization states are designated by the letters H for horizontal, and V, for vertical. Thus, there can be four combinations of both transmit and receive polarizations as follows:

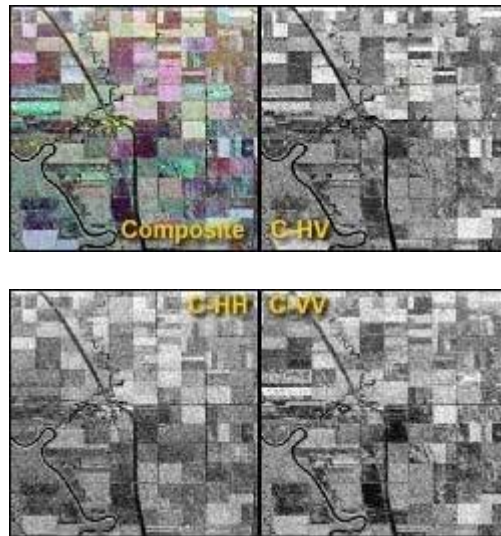
- HH - for horizontal transmit and horizontal receive,
- VV - for vertical transmit and vertical receive,
- HV - for horizontal transmit and vertical receive, and
- VH - for vertical transmit and horizontal receive.



The first two polarization combinations are referred to as like-polarized because the transmit and receive polarizations are the same. The last two combinations are referred to as cross-polarized because the transmit and receive polarizations are opposite of one another. These **C-band images** of agricultural fields demonstrate the variations in radar response due to changes in polarization. The bottom two images are like-polarized (HH and VV, respectively), and the upper right image is cross-polarized (HV). The upper left image is the result of displaying each of the three different polarizations together, one through each of the primary colours (red, green, and blue). Similar to variations in wavelength, depending on the transmit and receive polarizations, the radiation will interact with and be

Section 3.2 Radar Basics

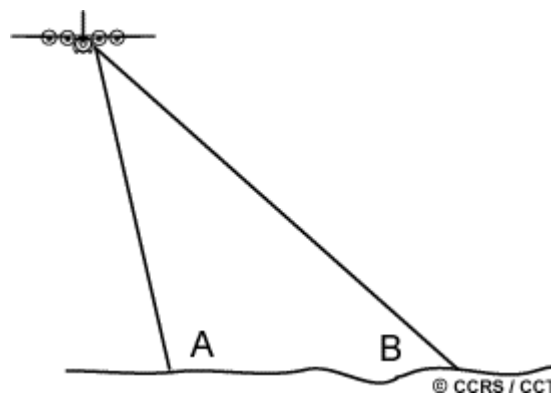
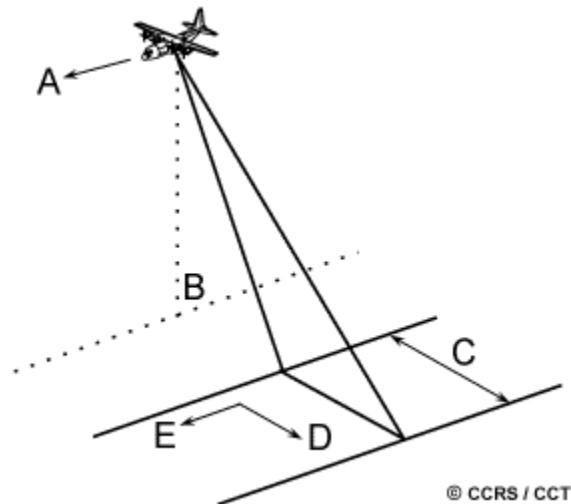
backscattered differently from the surface. Both wavelength and polarization affect how a radar "sees" the surface. Therefore, radar imagery collected using different polarization and wavelength combinations may provide different and complementary information about the targets on the surface.



C-band images

3.3 Viewing Geometry and Spatial Resolution

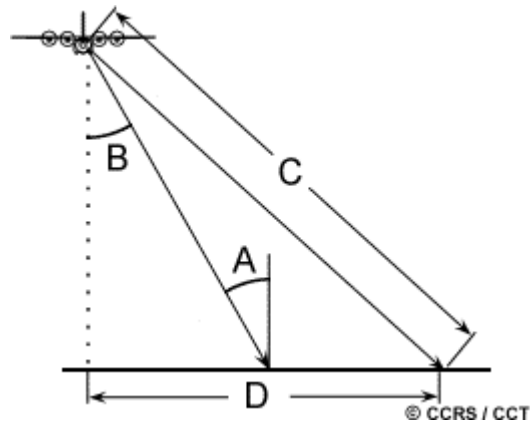
The imaging geometry of a radar system is different from the framing and scanning systems commonly employed for optical remote sensing described in Chapter 2. Similar to optical systems, the platform travels forward in the **flight direction (A)** with the **nadir (B)** directly beneath the platform. The microwave beam is transmitted obliquely at right angles to the direction of flight illuminating a **swath (C)** which is offset from nadir. **Range (D)** refers to the across-track dimension perpendicular to the flight direction, while **azimuth (E)** refers to the along-track dimension parallel to the flight direction. This side-looking viewing geometry is typical of imaging radar systems (airborne or spaceborne).



Near range

The portion of the image swath closest to the nadir track of the radar platform is called the **near range (A)** while the portion of the swath farthest from the nadir is called the **far range(B)**.

Section 3.3 Viewing Geometry and Spatial Resolution

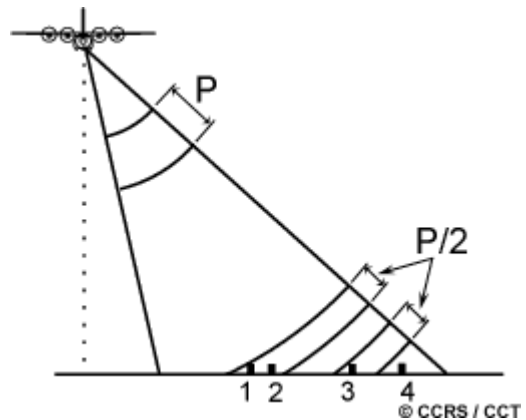


Incidence angle

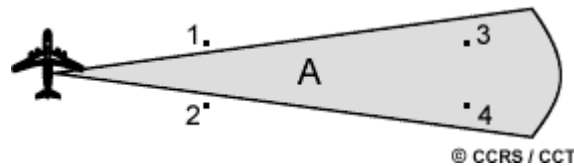
The incidence angle is the angle between the **radar beam and ground surface (A)** which increases, moving across the swath from near to far range. The **look angle (B)** is the angle at which the radar "looks" at the surface. In the near range, the viewing geometry may be referred to as being steep, relative to the far range, where the viewing geometry is shallow. At all ranges the radar antenna measures the radial line of sight distance between the radar and each target on the surface. This is the **slant range distance (C)**. The **ground range distance (D)** is the true horizontal distance along the ground corresponding to each point measured in slant range.

Unlike optical systems, a radar's spatial resolution is a function of the specific properties of the microwave radiation and geometrical effects. If a Real Aperture Radar (RAR) is used for image formation (as in Side-Looking Airborne Radar) a single transmit pulse and the backscattered signal are used to form the image. In this case, the resolution is dependent on the effective length of the pulse in the slant range direction and on the width of the illumination in the azimuth direction. The **range or across-track resolution** is dependent on the length of the pulse (P). Two

distinct targets on the surface will be resolved in the range dimension if their separation is greater than half the pulse length. For example, targets 1 and 2 will not be separable while targets 3 and 4 will. Slant range resolution remains constant, independent of range. However, when projected into ground range coordinates, the resolution in ground range will be dependent of the incidence angle. Thus, for fixed slant range resolution, the ground range resolution will decrease with increasing range.

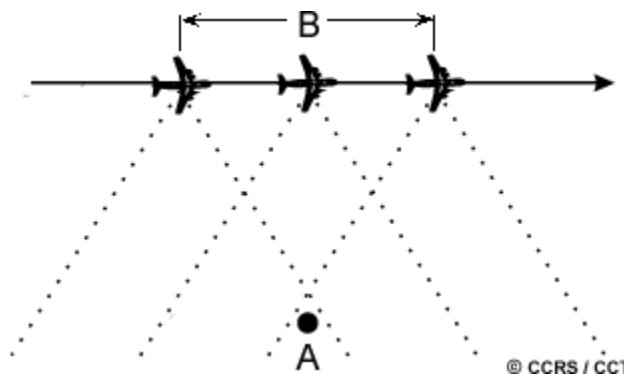


Section 3.3 Viewing Geometry and Spatial Resolution



The **azimuth or along-track resolution** is determined by the angular width of the radiated microwave beam and the slant range distance. This **beamwidth (A)** is a measure of the width of the illumination pattern. As the radar illumination propagates to increasing distance from the sensor, the azimuth resolution increases (becomes coarser). In this illustration, targets 1 and 2 in the near range would be separable, but targets 3 and 4 at further range would not. The radar beamwidth is inversely proportional to the antenna length (also referred to as the aperture) which means that a longer antenna (or aperture) will produce a narrower beam and finer resolution.

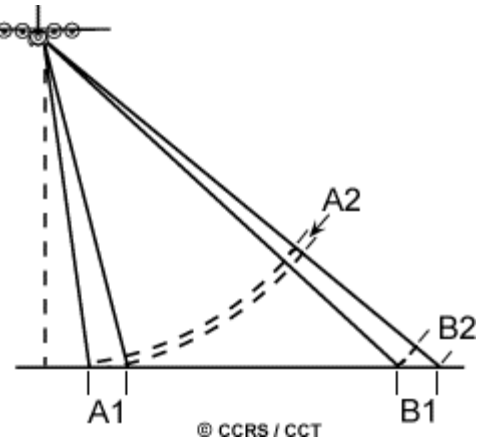
Finer range resolution can be achieved by using a shorter pulse length, which can be done within certain engineering design restrictions. Finer azimuth resolution can be achieved by increasing the antenna length. However, the actual length of the antenna is limited by what can be carried on an airborne or spaceborne platform. For airborne radars, antennas are usually limited to one to two metres; for satellites they can be 10 to 15 metres in length. To overcome this size limitation, the forward motion of the platform and special recording and processing of the backscattered echoes are used to simulate a very long antenna and thus **increase azimuth resolution**.



This figure illustrates how this is achieved. As a **target (A)** first enters the radar beam (1), the backscattered echoes from each transmitted pulse begin to be recorded. As the platform continues to move forward, all echoes from the target for each pulse are recorded during the entire time that the target is within the beam. The point at which the target leaves the view of the radar beam (2) some time later, determines the length of the simulated or **synthesized antenna (B)**. Targets at far range, where the beam is widest will be illuminated for a longer period of time than objects at near range. The expanding beamwidth, combined with the increased time a target is within the beam as ground range increases, balance each other, such that the resolution remains constant across the entire swath. This method of achieving uniform, fine azimuth resolution across the entire imaging swath is called **synthetic aperture radar**, or **SAR**. Most airborne and spaceborne radars employ this type of radar.

3.4 Radar Image Distortions

As with all remote sensing systems, the viewing geometry of a radar results in certain geometric distortions on the resultant imagery. However, there are key differences for radar imagery which are due to the side-looking viewing geometry, and the fact that the radar is fundamentally a distance measuring device (i.e. measuring range). **Slant-range scale distortion** occurs because the radar is measuring the distance to features in slant-range rather than the true horizontal distance along the ground. This results in a varying image scale, moving from near to far range. Although targets A1



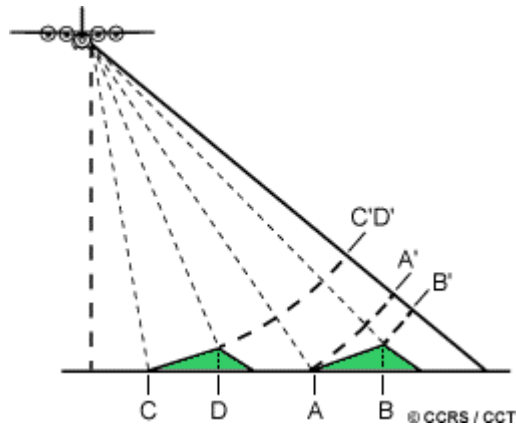
and B1 are the same size on the ground, their apparent dimensions in slant range (A2 and B2) are different. This causes targets in the near range to appear compressed relative to the far range. Using trigonometry, ground-range distance can be calculated from the slant-range distance and platform altitude to convert to the proper ground-range format.



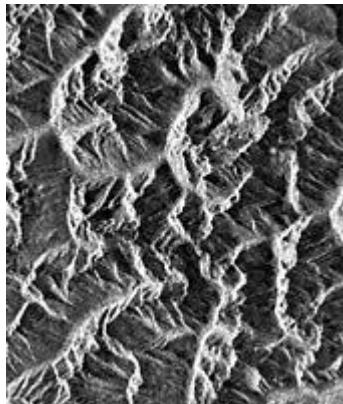
This conversion comparison shows a radar image in slant-range display (top) where the fields and the road in the near range on the left side of the image are compressed, and the same image converted to ground-range display (bottom) with the features in their proper geometric shape.

Similar to the distortions encountered when using cameras and scanners, radar images are also subject to geometric distortions due to **relief displacement**. As with scanner imagery, this displacement is one-dimensional and occurs perpendicular to the flight path. However, the displacement is reversed with targets being displaced towards, instead of away from the sensor. Radar **foreshortening** and **layover** are two consequences which result from relief displacement.

Section 3.4 Radar Image Distortions

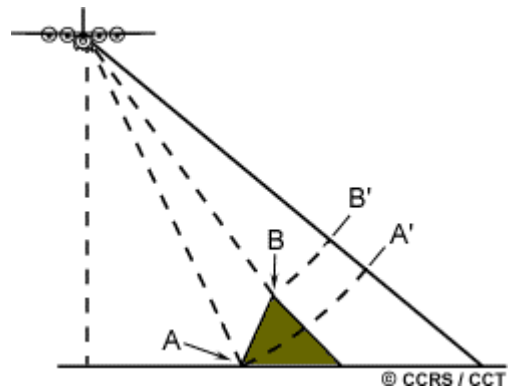


When the radar beam reaches the base of a tall feature tilted towards the radar (e.g. a mountain) before it reaches the top **foreshortening** will occur. Again, because the radar measures distance in slant-range, the slope (A to B) will appear compressed and the length of the slope will be represented incorrectly (A' to B'). Depending on the angle of the hillside or mountain slope in relation to the incidence angle of the radar beam, the severity of foreshortening will vary. Maximum foreshortening occurs when the radar beam is perpendicular to the slope such that the slope, the base, and the top are imaged simultaneously (C to D). The length of the slope will be reduced to an effective length of zero in slant range (C'D'). The figure below shows a radar image of **steep mountainous terrain** with severe foreshortening effects. The foreshortened slopes appear as bright features on the image.



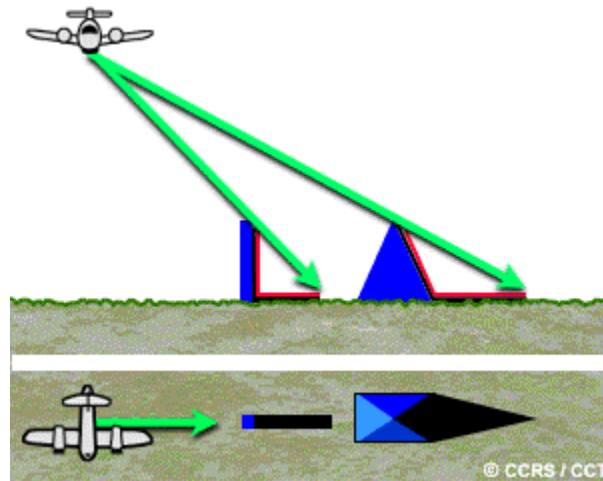
Section 3.4 Radar Image Distortions

Layover occurs when the radar beam reaches the top of a tall feature (B) before it reaches the base (A). The return signal from the top of the feature will be received before the signal from the bottom. As a result, the top of the feature is displaced towards the radar from its true position on the ground, and "lays over" the base of the feature (B' to A'). **Layover effects** on a radar image look very similar to effects due to foreshortening. As with foreshortening, layover is most severe for small incidence angles, at the near range of a swath, and in mountainous terrain.



Both foreshortening and layover result in **radar shadow**. Radar shadow occurs when the radar beam is not able to illuminate the ground surface. Shadows occur in the down range dimension (i.e. towards the far range), behind vertical features or slopes with steep sides. Since the radar beam does not illuminate the surface, shadowed regions will appear dark on an image as no energy is available to be backscattered. As incidence angle increases from near to far range, so will shadow effects as the radar beam looks more and more obliquely at the surface. This image illustrates **radar shadow effects** on the right side of the hillsides which are being illuminated from the left.

Section 3.4 Radar Image Distortions



Red surfaces are completely in shadow. Black areas in image are shadowed and contain no information.

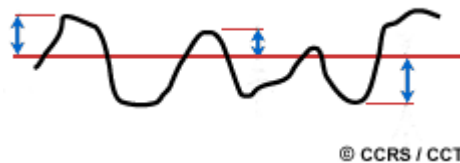


Radar shadow effects

3.5 Target Interaction and Image Appearance

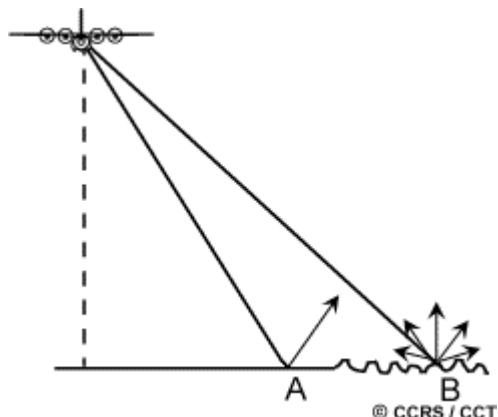
The brightness of features in a radar image is dependent on the portion of the transmitted energy that is returned back to the radar from targets on the surface. The magnitude or intensity of this backscattered energy is dependent on how the radar energy interacts with the surface, which is a function of several variables or parameters. These parameters include the particular characteristics of the radar system (frequency, polarization, viewing geometry, etc.) as well as the characteristics of the surface (landcover type, topography, relief, etc.). Because many of these characteristics are interrelated, it is impossible to separate out each of their individual contributions to the appearance of features in a radar image. Changes in the various parameters may have an impact on and affect the response of other parameters, which together will affect the amount of backscatter. Thus, the brightness of features in an image is usually a combination of several of these variables. However, for the purposes of our discussion, we can group these characteristics into three areas which fundamentally control radar energy/target interactions. They are:

- Surface roughness of the target
- Radar viewing and surface geometry relationship
- Moisture content and electrical properties of the target



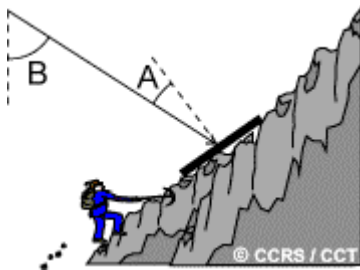
The surface roughness of a feature controls how the microwave energy interacts with that surface or target and is generally the dominant factor in determining the tones seen on a radar image. **Surface roughness** refers to the average height variations in the surface cover from a plane surface, and is measured on the order of centimetres. Whether a surface appears rough or smooth to a radar depends on the wavelength and incidence angle.

Simply put, a surface is considered "smooth" if the height variations are much smaller than the radar wavelength. When the surface height variations begin to approach the size of the wavelength, then the surface will appear "rough". Thus, a given surface will appear rougher as the wavelength becomes shorter and smoother as the wavelength becomes longer. A **smooth surface (A)** causes **specular** reflection of the incident energy (generally away from the sensor) and thus only a small amount of energy is returned to the radar. This results in smooth surfaces appearing as



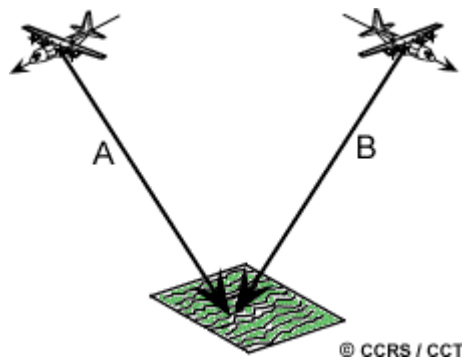
Section 3.5 Target Interaction and Image Appearance

darker toned areas on an image. A **rough surface (B)** will scatter the energy approximately equally in all directions (i.e. **diffusely**) and a significant portion of the energy will be backscattered to the radar. Thus, rough surfaces will appear lighter in tone on an image. Incidence angle, in combination with wavelength, also plays a role in the apparent roughness of a surface. For a given surface and wavelength, the surface will appear smoother as the incidence angle increases. Thus, as we move farther across the swath, from near to far range, less energy would be returned to the sensor and the image would become increasingly darker in tone.



We have already discussed incidence or look angle in relation to viewing geometry and how changes in this angle affect the signal returned to the radar. However, in relation to surface geometry, and its effect on target interaction and image appearance, the local incidence angle is a more appropriate and relevant concept. The local incidence angle is the angle between the radar beam and a line perpendicular to the slope at the point of incidence (A). Thus, local incidence angle takes into account the local slope of the terrain in relation to the radar beam. With flat terrain, the local incidence angle is the same as the look angle (B) of the radar. For terrain with any type of relief, this is not the case. Generally, slopes facing towards the radar will have small local incidence angles, causing relatively strong backscattering to the sensor, which results in a bright-toned appearance in an image.

As the concept of **local incidence angle** demonstrates, the relationship between viewing geometry and the geometry of the surface features plays an important role in how the radar energy interacts with targets and their corresponding brightness on an image. Variations in viewing geometry will accentuate and enhance topography and relief in different ways, such that varying degrees of foreshortening, layover, and shadow (section 3.4) may occur depending on surface slope, orientation, and shape.

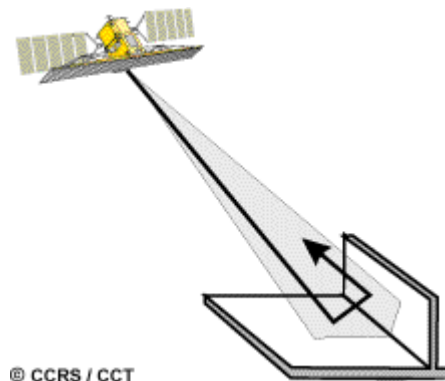


The **look direction or aspect angle** of the radar describes the orientation of the transmitted radar beam relative to the direction or alignment of linear features on the surface. The look direction can significantly influence the appearance of features on a radar image, particularly when ground features are organized in a linear structure (such as agricultural crops or

Section 3.5 Target Interaction and Image Appearance

mountain ranges). If the look direction is close to perpendicular to the orientation of the feature (A), then a large portion of the incident energy will be reflected back to the sensor and the feature will appear as a brighter tone. If the look direction is more oblique in relation to the feature orientation (B), then less energy will be returned to the radar and the feature will appear darker in tone. Look direction is important for enhancing the contrast between features in an image. It is particularly important to have the proper look direction in mountainous regions in order to minimize effects such as layover and shadowing. By acquiring imagery from different look directions, it may be possible to enhance identification of features with different orientations relative to the radar.

Features which have two (or more) surfaces (usually smooth) at right angles to one another, may cause **corner reflection** to occur if the 'corner' faces the general direction of the radar antenna. The orientation of the surfaces at right angles causes most of the radar energy to be reflected directly back to the antenna due to the double bounce (or more) reflection. Corner reflectors with complex angular shapes are common in urban environments (e.g. buildings and streets, bridges, other man-made structures). Naturally occurring corner reflectors may include severely folded rock and cliff



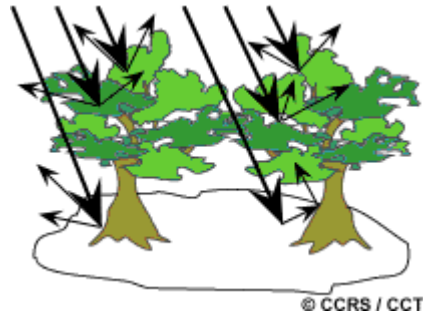
faces or upright vegetation standing in water. In all cases, corner reflectors show up as very bright targets in an image, such as the buildings and other man-made structures in this **radar image of a city**.



The presence (or absence) of moisture affects the electrical properties of an object or medium. Changes in the electrical properties influence the absorption, transmission, and reflection of microwave energy. Thus, the moisture content will influence how targets and surfaces reflect energy from a radar and how they will appear on an image. Generally, reflectivity (and image brightness) increases with increased moisture content. For example, surfaces such as soil and vegetation cover will appear brighter when they are wet than when they are dry.

Section 3.5 Target Interaction and Image Appearance

When a target is moist or wet, scattering from the topmost portion (surface scattering) is the dominant process taking place. The type of reflection (ranging from specular to diffuse) and the magnitude will depend on how rough the material appears to the radar. If the target is very dry and the surface appears smooth to the radar, the radar energy may be able to penetrate below the surface, whether that surface is discontinuous (e.g. forest canopy with leaves and branches), or a homogeneous surface (e.g. soil, sand, or ice). For a given surface, longer wavelengths are able to penetrate further than shorter wavelengths.

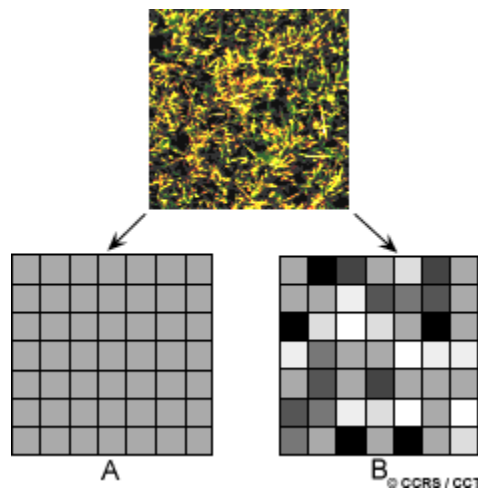


If the radar energy does manage to penetrate through the topmost surface, then volume scattering may occur. **Volume scattering** is the scattering of radar energy within a volume or medium, and usually consists of multiple bounces and reflections from different components within the volume. For example, in a forest, scattering may come from the leaf canopy at the tops of the trees, the leaves and branches further below, and the tree trunks and soil at the ground level. Volume scattering may serve to decrease or increase image brightness, depending on how much of the energy is scattered out of the volume and back to the radar.

3.6 Radar Image Properties



All radar images appear with some degree of what we call radar speckle. **Speckle** appears as a grainy "salt and pepper" texture in an image. This is caused by random constructive and destructive interference from the multiple scattering returns that will occur within each resolution cell. As an example, an homogeneous target, such as a large grass-covered field, without the effects of speckle would generally result in light-toned pixel values on an image (A). However, reflections from the individual blades of grass within each resolution cell results in some image pixels being brighter and some being darker than the average tone (B), such that the field appears speckled.

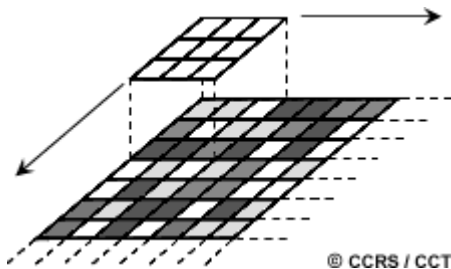
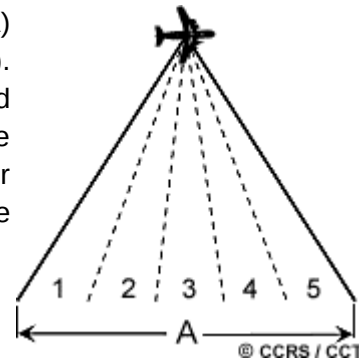


Speckle is essentially a form of noise which degrades the quality of an image and may make interpretation (visual or digital) more difficult. Thus, it is generally desirable to reduce speckle prior to interpretation and analysis. **Speckle reduction** can be achieved in two ways:

- multi-look processing, or
- spatial filtering.

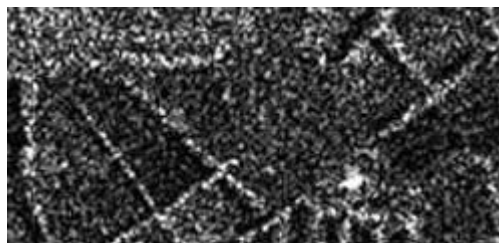
Section 3.6 Radar Image Properties

Multi-look processing refers to the division of the radar beam (A) into several (in this example, five) narrower sub-beams (1 to 5). Each sub-beam provides an independent "look" at the illuminated scene, as the name suggests. Each of these "looks" will also be subject to speckle, but by summing and averaging them together to form the final output image, the amount of speckle will be reduced.



While multi-look processing is usually done during data acquisition, speckle reduction by spatial filtering is performed on the output image in a digital (i.e. computer) image analysis environment. Speckle reduction filtering consists of moving a small window of a few pixels in dimension (e.g. 3x3 or 5x5) over each pixel in the image, applying a mathematical calculation using the pixel

values under that window (e.g. calculating the average), and replacing the central pixel with the new value. The window is moved along in both the row and column dimensions one pixel at a time, until the entire image has been covered. By calculating the average of a small window around each pixel, a smoothing effect is achieved and the visual appearance of the speckle is reduced.



Speckle reduction using an averaging filter

This graphic shows a radar image before (top) and after (bottom) speckle reduction using an averaging filter. The median (or middle) value of all the pixels underneath the moving window is also often used to reduce speckle. Other more complex filtering calculations can be performed to reduce speckle while minimizing the amount of smoothing taking place.

Section 3.6 Radar Image Properties

Both multi-look processing and spatial filtering reduce speckle at the expense of resolution, since they both essentially smooth the image. Therefore, the amount of speckle reduction desired must be balanced with the particular application the image is being used for, and the amount of detail required. If fine detail and high resolution is required then little or no multi-look/spatial filtering should be done. If broad-scale interpretation and mapping is the application, then speckle reduction techniques may be more appropriate and acceptable.

Another property peculiar to radar images is slant-range distortion, which was discussed in some detail in section 3.4. Features in the near-range are compressed relative to features in the far range due to the slant-range scale variability. For most applications, it is desirable to have the radar image presented in a format which corrects for this distortion, to enable true distance measurements between features. This requires the slant-range image to be converted to 'ground range' display. This can be done by the radar processor prior to creating an image or after data acquisition by applying a transformation to the slant range image. In most cases, this conversion will only be an estimate of the geometry of the ground features due to the complications introduced by variations in terrain relief and topography.

A radar antenna transmits more power in the mid-range portion of the illuminated swath than at the near and far ranges. This effect is known as **antenna pattern** and results in stronger returns from the center portion of the swath than at the edges. Combined with this antenna pattern effect is the fact that the energy returned to the radar decreases dramatically as the range distance increases. Thus, for a given surface, the strength of the returned signal becomes smaller and smaller moving farther across the swath. These effects combine to produce an image which varies in intensity (tone) in the range direction across the image. A process known as **antenna pattern correction** may be applied to produce a uniform average brightness across the imaged swath, to better facilitate visual interpretation.



The range of brightness levels a remote sensing system can differentiate is related to radiometric resolution (section 2.5) and is referred to as the **dynamic range**. While optical sensors, such as those carried by satellites such as Landsat and SPOT, typically produce 256 intensity levels, radar systems can differentiate intensity levels up to around 100,000 levels! Since the human eye can only discriminate about 40 intensity levels at one time, this is too much information for visual interpretation. Even a typical computer would have difficulty dealing with this range of information. Therefore, most radars record and process the original

Section 3.6 Radar Image Properties

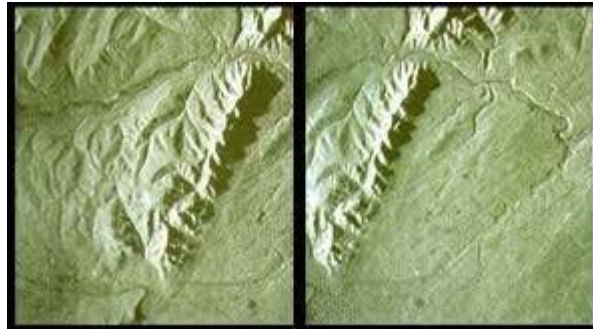
data as 16 bits (65,536 levels of intensity), which are then further scaled down to 8 bits (256 levels) for visual interpretation and/or digital computer analysis.

Calibration is a process which ensures that the radar system and the signals that it measures are as consistent and as accurate as possible. Prior to analysis, most radar images will require **relative calibration**. Relative calibration corrects for known variations in radar antenna and systems response and ensures that uniform, repeatable measurements can be made over time. This allows relative comparisons between the response of features within a single image, and between separate images to be made with confidence. However, if we wish to make accurate **quantitative** measurements representing the actual energy or power returned from various features or targets for comparative purposes, then **absolute calibration** is necessary.

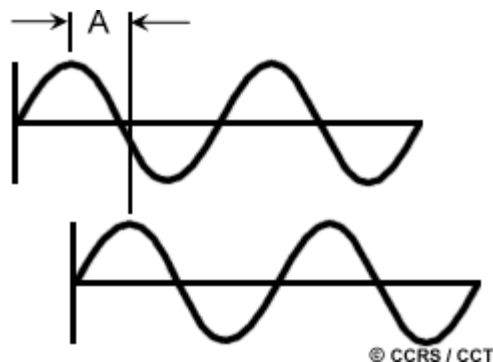
Absolute calibration, a much more involved process than relative calibration, attempts to relate the magnitude of the recorded signal strength to the actual amount of energy backscattered from each resolution cell. To achieve this, detailed measurements of the radar system properties are required as well as quantitative measurements of the scattering properties of specific targets. The latter are often obtained using ground-based scatterometers, as described in section 3.1. Also, devices called **transponders** may be placed on the ground prior to data acquisition to calibrate an image. These devices receive the incoming radar signal, amplify it, and transmit a return signal of known strength back to the radar. By knowing the actual strength of this return signal in the image, the responses from other features can be referenced to it.

3.7 Advanced Radar Applications

In addition to standard acquisition and use of radar data, there are three specific applications worth mentioning.



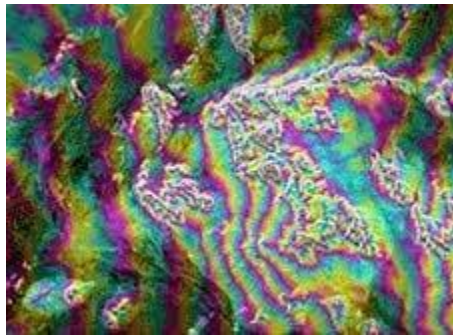
The first is **stereo radar** which is similar in concept to stereo mapping using aerial photography (described in section 2.7). Stereo radar image pairs are acquired covering the same area, but with different look/incidence angles (A), or opposite look directions (B). Unlike aerial photos where the displacement is radially outward from the nadir point directly below the camera, radar images show displacement only in the range direction. Stereo pairs taken from opposite look directions (i.e. one looking north and the other south) may show significant contrast and may be difficult to interpret visually or digitally. In mountainous terrain, this will be even more pronounced as shadowing on opposite sides of features will eliminate the stereo effect. Same side stereo imaging (A) has been used operationally for years to assist in interpretation for forestry and geology and also to generate topographic maps. The estimation of distance measurements and terrain height for topographic mapping from stereo radar data is called **radargrammetry**, and is analogous to photogrammetry carried out for similar purposes with aerial photographs.



Radargrammetry is one method of estimating terrain height using radar. Another, more advanced method is called **interferometry**. Interferometry relies on being able to measure a property of electromagnetic waves called **phase**. Suppose we have **two waves** with the exact

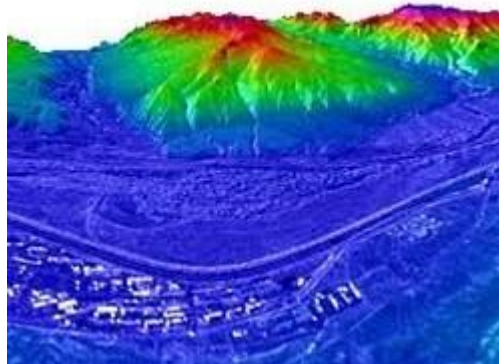
Section 3.7 Advanced Radar Applications

same wavelength and frequency traveling along in space, but the starting point of one is offset slightly from the other. The offset between matching points on these two waves (A) is called the **phase difference**. **Interferometric systems** use two antennas, separated in the range dimension by a small distance, both recording the returns from each resolution cell. The two antennas can be on the same platform (as with some airborne SARs), or the data can be acquired from two different passes with the same sensor, such has been done with both airborne and satellite radars. By measuring



the exact phase difference between the two returns (A), the path length difference can be calculated to an accuracy that is on the order of the wavelength (i.e centimetres). Knowing the position of the antennas with respect to the Earth's surface, the position of the resolution cell, including its elevation, can be determined. The phase difference between adjacent resolution cells, is illustrated in this **interferogram**, where colours

represents the variations in height. The information contained in an interferogram can be used to derive topographic information and produce **three-dimensional imagery** of terrain height.



The concept of radar **polarimetry** was already alluded to in our discussion of radar fundamentals in section 3.2. As its name implies, polarimetry involves discriminating between the **polarizations** that a radar system is able to transmit and receive. Most radars transmit microwave radiation in either horizontal (H) or vertical (V) polarization, and similarly, receive the backscattered signal at only one of these polarizations. **Multi-polarization** radars are able to transmit either H or V polarization and receive both the like- and cross-polarized returns (e.g. HH and HV or VV and VH, where the first letter stands for the polarization transmitted and the second letter the polarization received). **Polarimetric radars** are able to transmit and receive both horizontal and vertical polarizations. Thus, they are able to receive and process all four combinations of these polarizations: HH, HV, VH, and VV. Each of these "polarization channels" have varying sensitivities to different surface characteristics and properties. Thus,

Section 3.7 Advanced Radar Applications

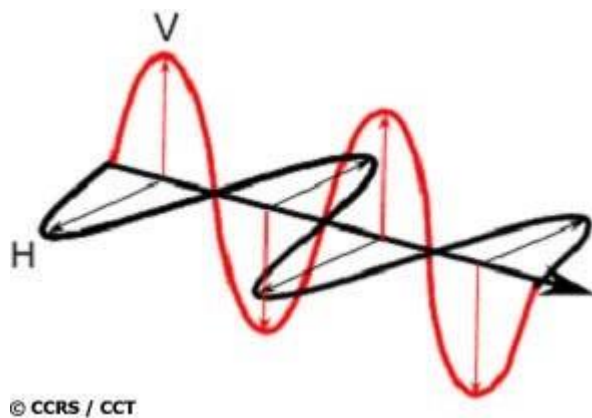
the availability of multi-polarization data helps to improve the identification of, and the discrimination between features. In addition to recording the magnitude (i.e. the strength) of the returned signal for each polarization, most polarimetric radars are also able to record the **phase** information of the returned signals. This can be used to further characterize the polarimetric "signature" of different surface features.

3.8 Radar Polarimetry

Introduction to Polarization

When discussing microwave energy propagation and scattering, the polarization of the radiation is an important property. For a plane electromagnetic (EM) wave, polarization refers to the locus of the electric field vector in the plane perpendicular to the direction of propagation. While the length of the vector represents the **amplitude** of the wave, and the rotation rate of the vector represents the **frequency** of the wave, polarization refers to the **orientation** and **shape** of the pattern traced by the tip of the vector.

The waveform of the electric field strength (voltage) of an EM wave can be predictable (the wave is polarized) or random (the wave is unpolarized), or a combination of both. In the latter case, the degree of polarization describes the ratio of polarized power to total power of the wave. An example of a fully polarized wave would be a **monochromatic** sine wave, with a single, constant frequency and stable amplitude.



Examples of horizontal (black) and vertical (red) polarizations of a plane electromagnetic wave

Many radars are designed to transmit microwave radiation that is either horizontally polarized (H) or vertically polarized (V). A transmitted wave of either polarization can generate a backscattered wave with a variety of polarizations. It is the analysis of these transmit and receive polarization combinations that constitutes the science of radar polarimetry.

Any polarization on either transmission or reception can be synthesized by using H and V components with a well-defined relationship between them. For this reason, systems that transmit and receive both of these linear polarizations are commonly used. With these radars, there can be four combinations of transmit and receive polarizations:

- HH - for horizontal transmit and horizontal receive
- VV - for vertical transmit and vertical receive

Section 3.8 Radar Polarimetry

- HV - for horizontal transmit and vertical receive, and
- VH - for vertical transmit and horizontal receive.

The first two polarization combinations are referred to as "like-polarized" because the transmit and receive polarizations **are the same**. The last two combinations are referred to as "cross-polarized" because the transmit and receive polarizations **are orthogonal** to one another.

Radar systems can have one, two or all four of these transmit/receive polarization combinations. Examples include the following types of radar systems:

single polarized	- HH or VV (or possibly HV or VH)
dual polarized	- HH and HV, VV and VH, or HH and VV
alternating polarization	- HH and HV, alternating with VV and VH
polarimetric	- HH, VV, HV, and VH

Note that "quadrature polarization" and "fully polarimetric" can be used as synonyms for "polarimetric". The **relative phase** between channels is measured in a polarimetric radar, and is a very important component of the measurement. In the other radar types, relative phase may or may not be measured. The alternating polarization mode has been introduced on ENVISAT - relative phase is measured but the important HH-VV phase is not meaningful because of the time lapse between the measurements.

These C-band images of agricultural fields demonstrate the dependence of the radar response on polarization. The top two images are like-polarized (HH on left, VV on right), and the lower left image is cross-polarized (HV). The lower right image is the result of displaying these three images as a colour composite (in this case, HH - red, VV - green, and HV - blue).

Both wavelength and polarization affect how a radar system "sees" the elements in the scene. Therefore, radar imagery collected using different polarization and wavelength combinations may provide different and complementary information. Furthermore, when three polarizations are combined in a colour composite, the information is presented in a way that an image interpreter can infer more information of the surface characteristics.

Section 3.8 Radar Polarimetry

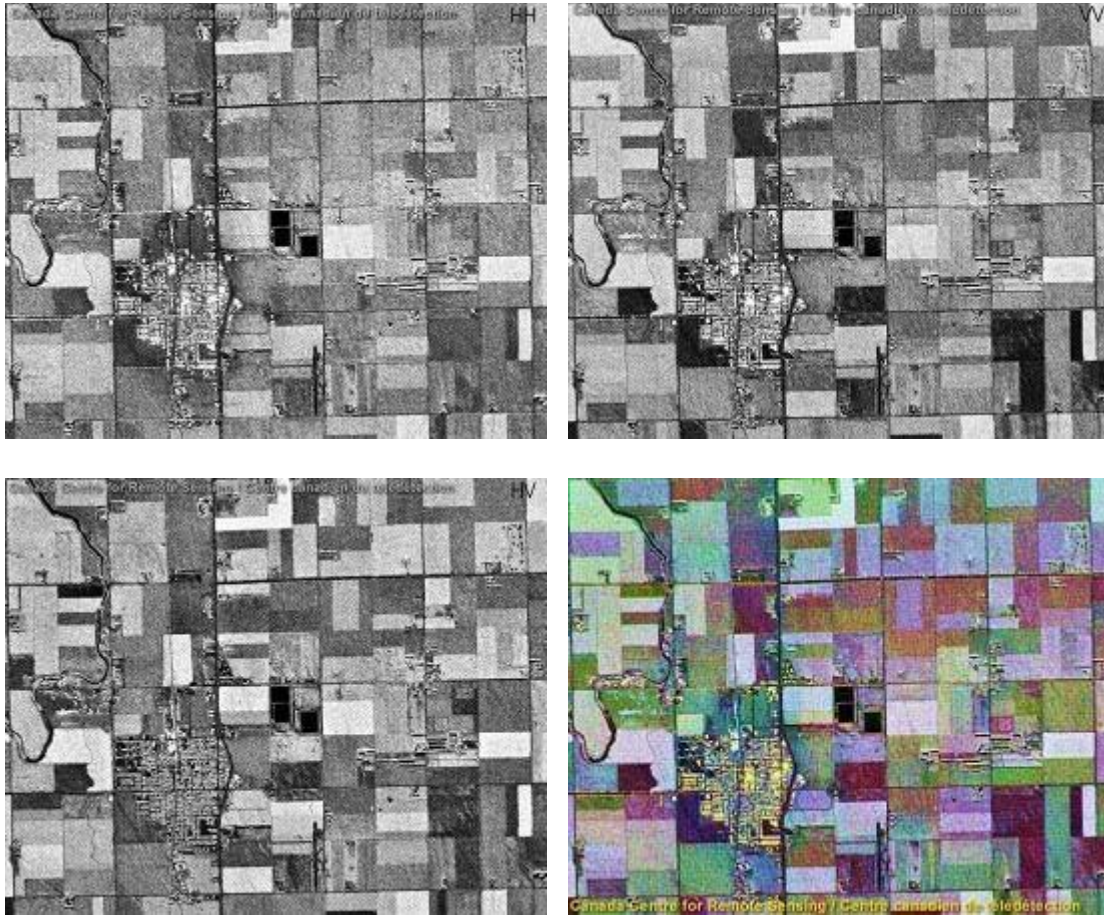


Illustration of how different polarizations (HH, VV, HV & colour composite) bring out different features in an agricultural scene

Polarimetric Information

The primary description of how a radar target or surface feature scatters EM energy is given by the scattering matrix. From the scattering matrix, other forms of polarimetric information can be derived, such as synthesized images and polarization signatures.

Polarization Synthesis

A polarimetric radar can be used to determine the target response or scattering matrix using two orthogonal polarizations, typically linear H and linear V on each of transmit and receive. If a scattering matrix is known, the response of the target to **any combination** of incident and received polarizations can be computed. This is referred to as **polarization synthesis**, and illustrates the power and flexibility of a fully polarimetric radar.

Through polarization synthesis, an image can be created to improve the detectability of

Section 3.8 Radar Polarimetry

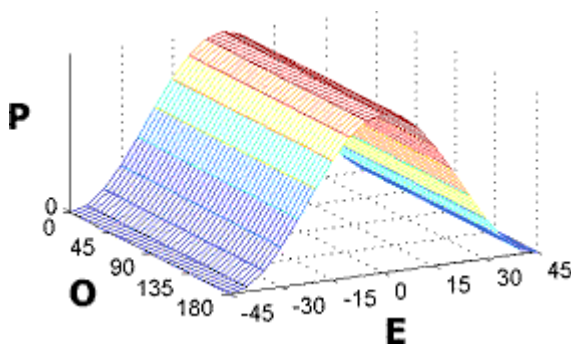
selected features. An example is the detection of ships in ocean images. To find the best transmit-receive polarization combination to use, the polarization signature of a typical ship and that of the ocean is calculated for a number of polarizations. Then the ratio of the ship to ocean backscatter is computed for each polarization. The transmit-receive polarization combination that maximises the ratio of backscatter strength is then used to improve the detectability of ships. This procedure is called "polarimetric contrast enhancement" or the use of a "polarimetric matched filter".

Polarization Signatures

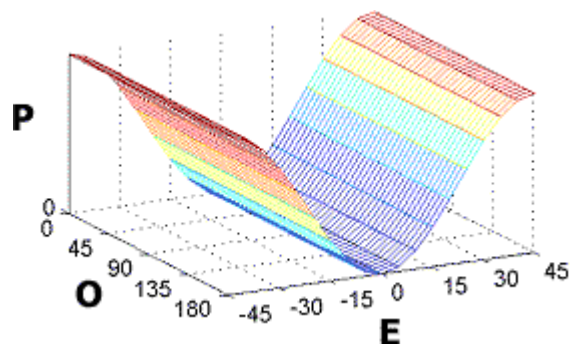
Because the incident and scattered waves can take on so many different polarizations, and the scattering matrix consists of four complex numbers, it is helpful to simplify the interpretation of the scattering behaviour using three-dimensional plots. The "polarization signature" of the target provides a convenient way of visualising a target's scattering properties. The signatures are also called "polarization response plots".

An incident electromagnetic wave can be selected to have an electric field with ellipticity between -45° and $+45^\circ$, and an orientation between 0 and 180° . These variables are used as the x- and y-axes of a 3-D plot portraying the polarization signature. For each of these possible incident polarizations, the strength of the backscatter can be computed for the **same** polarization on transmit and receive (the co-polarized signature) and for **orthogonal** polarizations on transmit and receive (the cross-polarized signature). The strength is displayed on the z-axis of the signatures.

Co-polarized signature



Cross-polarized signature



Polarization signatures of a large conducting sphere.

P = Power, O = Orientation (degrees), E = Ellipticity (degrees)

This figure shows the polarization signatures of the most simple of all targets - a large conducting sphere or a trihedral corner reflector. The wave is backscattered with the same polarization, except for a change of sign of the ellipticity (or in the case of linear polarization, a

Section 3.8 Radar Polarimetry

change of the phase angle between E_h and E_v of 180°). The sign changes once for every reflection - the sphere represents a single reflection, and the trihedral gives three reflections, so each behaves as an "odd-bounce" reflector.

For more complicated targets, the polarization signature takes on different shapes. Two interesting signatures come from a dihedral corner reflector and Bragg scattering from the sea surface. In the case of the dihedral reflector, the co-pol signature has a double peak, characteristic of "even-bounce" reflectors. In the case of Bragg scattering, the response is similar to the single-bounce sphere, except that the backscatter of the vertical polarization is higher than that of the horizontal polarization.

Data Calibration

One critical requirement of polarimetric radar systems is the need for calibration. This is because much of the information lies in the ratios of amplitudes and the differences in phase angle between the four transmit-receive polarization combinations. If the calibration is not sufficiently accurate, the scattering mechanisms will be misinterpreted and the advantages of using polarization will not be realised.

Calibration is achieved by a combination of radar system design and data analysis. Imagine the response to a trihedral corner reflector. Its ideal response is only obtained if the four channels of the radar system all have the same gain, system-dependent phase differences between channels are absent, and there is no energy leakage from one channel to another.

In terms of the radar system design, the channel gains and phases should be as carefully matched as possible. In the case of the phase balance, this means that the signal path lengths should be effectively the same in all channels. Calibration signals are often built into the design to help verify these channel balances.

In terms of data analysis, channel balances, cross-talk and noise effects can be measured and corrected by analysing the received data. In addition to analysing the response of internal calibration signals, the signals from known targets such as corner reflectors, active transponders, and uniform clutter can be used to calibrate some of the parameters.

Polarimetric Applications

Synthetic Aperture Radar polarimetry has been limited to a number of experimental airborne SAR systems and the SIR-C (shuttle) mission. With these data, researchers have studied a number of applications, and have shown that the interpretation of a number of features in a scene is facilitated when the radar is operated in polarimetric mode. The launch of RADARSAT-2 will make polarimetric data available on an operational basis, and uses of such data will become more routine and more sophisticated.

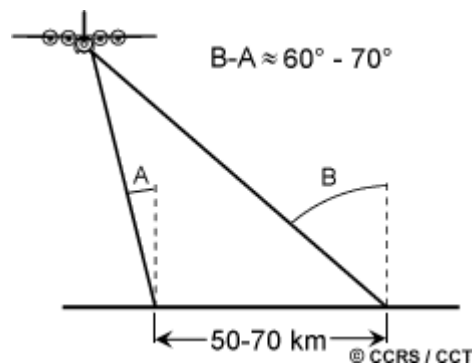
Some applications in which polarimetric SAR has already proved useful include:

Section 3.8 Radar Polarimetry

- Agriculture: for crop type identification, crop condition monitoring, soil moisture measurement, and soil tillage and crop residue identification;
- Forestry: for clearcuts and linear features mapping, biomass estimation, species identification and fire scar mapping;
- Geology: for geological mapping;
- Hydrology: for monitoring wetlands and snow cover;
- Oceanography: for sea ice identification, coastal windfield measurement, and wave slope measurement;
- Shipping: for ship detection and classification;
- Coastal Zone: for shoreline detection, substrate mapping, slick detection and general vegetation mapping.

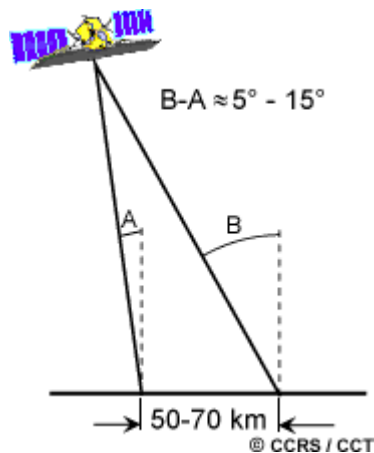
3.9 Airborne versus Spaceborne Radars

Like other remote sensing systems, an imaging radar sensor may be carried on either an airborne or spaceborne platform. Depending on the use of the prospective imagery, there are trade-offs between the two types of platforms. Regardless of the platform used, a significant advantage of using a Synthetic Aperture Radar (SAR) is that the spatial resolution is independent of platform altitude. Thus, fine resolution can be achieved from both airborne and spaceborne platforms.



Although spatial resolution is independent of altitude, viewing geometry and swath coverage can be greatly affected by altitude variations. At aircraft operating altitudes, an airborne radar must image over a wide range of incidence angles, perhaps as much as 60 or 70 degrees, in order to achieve relatively wide swaths (let's say 50 to 70 km). As we have learned in the preceding sections, incidence angle (or look angle) has a significant effect on the backscatter from surface features and on their appearance on an image. Image characteristics such as foreshortening, layover, and shadowing will be subject to wide variations, across a large incidence angle range. Spaceborne radars are able to avoid some of these imaging geometry problems since they operate at altitudes up to one hundred times higher than airborne radars. At altitudes of several hundred kilometres, spaceborne radars can image comparable swath widths, but over a much narrower range of incidence angles, typically ranging from five to 15 degrees. This provides for more uniform illumination and reduces undesirable imaging variations across the swath due to viewing geometry.

Section 3.9 Airborne versus Spaceborne Radars



Although airborne radar systems may be more susceptible to imaging geometry problems, they are flexible in their capability to collect data from different look angles and look directions. By optimizing the geometry for the particular terrain being imaged, or by acquiring imagery from more than one look direction, some of these effects may be reduced. Additionally, an airborne radar is able to collect data anywhere and at any time (as long as weather and flying conditions are acceptable!). A spaceborne radar does not have this degree of flexibility, as its viewing geometry and data acquisition schedule is controlled by the pattern of its orbit. However, satellite radars do have the advantage of being able to collect imagery more quickly over a larger area than an airborne radar, and provide consistent viewing geometry. The frequency of coverage may not be as often as that possible with an airborne platform, but depending on the orbit parameters, the viewing geometry flexibility, and the geographic area of interest, a spaceborne radar may have a revisit period as short as one day.

As with any aircraft, an airborne radar will be susceptible to variations in velocity and other motions of the aircraft as well as to environmental (weather) conditions. In order to avoid image artifacts or geometric positioning errors due to random variations in the motion of the aircraft, the radar system must use sophisticated navigation/positioning equipment and advanced image processing to compensate for these variations. Generally, this will be able to correct for all but the most severe variations in motion, such as significant air turbulence. Spaceborne radars are not affected by motion of this type. Indeed, the geometry of their orbits is usually very stable and their positions can be accurately calculated. However, geometric correction of imagery from spaceborne platforms must take into account other factors, such as the rotation and curvature of the Earth, to achieve proper geometric positioning of features on the surface.

3.10 Airborne and Spaceborne Radar Systems

In order to more clearly illustrate the differences between airborne and spaceborne radars, we will briefly outline a few of the representative systems of each type, starting with airborne systems.



The **Convair-580 C/X SAR** system developed and operated by the Canada Centre for Remote Sensing was a workhorse for experimental research into advanced SAR applications in Canada and around the world, particularly in preparation for satellite-borne SARs. The system was transferred to Environment Canada in 1996 for use in oil spill research and other environmental applications. This system operates at two radar bands, C- (5.66 cm) and X- (3.24 cm). Cross-polarization data can be recorded simultaneously for both the C- and X-band channels, and the C-band system can be operated as a fully polarimetric radar. Imagery can be acquired at three different imaging geometries (nadir, narrow and wide swath modes) over a wide range of incidence angles (five degrees to almost 90 degrees). In addition to being a fully calibratable system for quantitative measurements, the system has a second antenna mounted on the aircraft fuselage to allow the C-band system to be operated as an interferometric radar.



The **Sea Ice and Terrain Assessment (STAR)** systems operated by Intera Technologies Limited of Calgary, Alberta, Canada, (later Intermap Technologies) were among the first SAR systems used commercially around the world. Both STAR-1 and STAR-2 operate at X-band (3.2 cm) with HH polarization in two different resolution modes. The swath coverage varies from 19 to 50 km, and the resolution from 5 to 18 m. They were primarily designed for monitoring sea ice (one of the key applications for radar, in Canada) and for terrain

analysis. Radar's all-weather, day or night imaging capabilities are well-suited to monitoring ice in Canada's northern and coastal waters. STAR-1 was also the first SAR system to use on-board data processing and to offer real-time downlinking of data to surface stations.

Section 3.10 Airborne and Spaceborne Radar Systems

The United States National Aeronautics and Space Administration (NASA) has been at the forefront of multi-frequency, multi-polarization synthetic aperture radar research for many years. The Jet Propulsion Laboratory (JPL) in California has operated various advanced systems on contract for NASA. The **AirSAR**



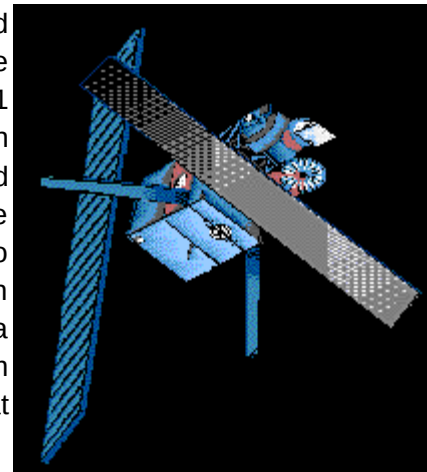
system is a C-, L-, and P-band advanced polarimetric SAR which can collect data for each of these bands at all possible combinations of horizontal and vertical transmit and receive polarizations (i.e. HH, HV, VH, and VV). Data from the AirSAR system can be fully calibrated to allow extraction of quantitative measurements of radar backscatter. Spatial resolution of the AirSAR system is on the order of 12 metres in both range and azimuth. Incidence angle ranges from zero degrees at nadir to about 70 degrees at the far range. This capability to collect multi-frequency, multi-polarization data over such a diverse range of incidence angles allows a wide variety of specialized research experiments to be carried out.



With the advances and success of airborne imaging radar, satellite radars were the next logical step to complement the optical satellite sensors in operation. **SEASAT**, launched in 1978, was the first civilian remote sensing satellite to carry a spaceborne SAR sensor. The SAR operated at L-band (23.5 cm) with HH polarization. The viewing geometry was fixed between nine and 15 degrees with a swath width of 100 km and a spatial resolution of 25 metres. This steep viewing geometry was designed primarily for

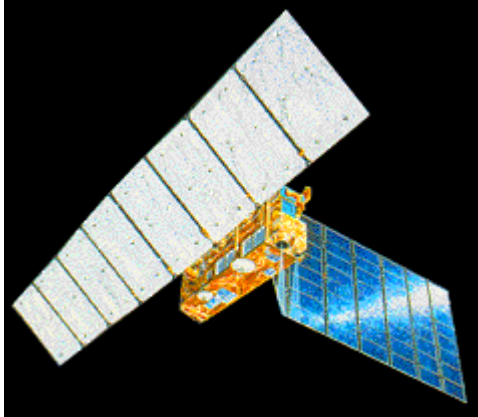
observations of ocean and sea ice, but a great deal of imagery was also collected over land areas. However, the small incidence angles amplified foreshortening and layover effects over terrain with high relief, limiting its utility in these areas. Although the satellite was only operational for three months, it demonstrated the wealth of information (and the large volumes of data!) possible from a spaceborne radar.

With the success of the short-lived SEASAT mission, and impetus provided from positive results with several airborne SARs, the European Space Agency (ESA) launched ERS-1 in July of 1991. **ERS-1** carried on-board a radar altimeter, an infrared radiometer and microwave sounder, and a C-band (5.66 cm), active microwave instrument. This is a flexible instrument which can be operated as a scatterometer to measure reflectivity of the ocean surface, as well as ocean surface wind speed and direction. It can also operate as a synthetic aperture radar, collecting imagery over a 100 km swath over an incidence angle range of 20 to 26 degrees, at a resolution of approximately 30 metres. Polarization is



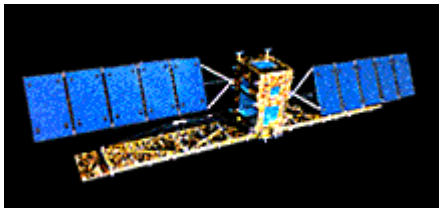
Section 3.10 Airborne and Spaceborne Radar Systems

vertical transmit and vertical receive (VV) which, combined with the fairly steep viewing angles, make ERS-1 particularly sensitive to surface roughness. The revisit period (or repeat cycle) of ERS-1 can be varied by adjusting the orbit, and has ranged from three to 168 days, depending on the mode of operation. Generally, the repeat cycle is about 35 days. A second satellite, ERS-2, was launched in April of 1995 and carries the same active microwave sensor as ERS-1. Designed primarily for ocean monitoring applications and research, ERS-1 provided the worldwide remote sensing community with the first wide-spread access to spaceborne SAR data. Imagery from both satellites has been used in a wide range of applications, over both ocean and land environments. Like SEASAT, the steep viewing angles limit their utility for some land applications due to geometry effects.



The National Space Development Agency of Japan (NASDA), launched the **JERS-1** satellite in February of 1992. In addition to carrying two optical sensors, JERS-1 has an L-band (23.5 cm) SAR operating at HH polarization. The swath width is approximately 75 km and spatial resolution is approximately 18 metres in both range and azimuth. The imaging geometry of JERS-1 is slightly shallower than either SEASAT or the ERS satellites, with the incidence angle at the middle of the swath being 35 degrees. Thus, JERS-1 images are slightly less susceptible to geometry and terrain effects.

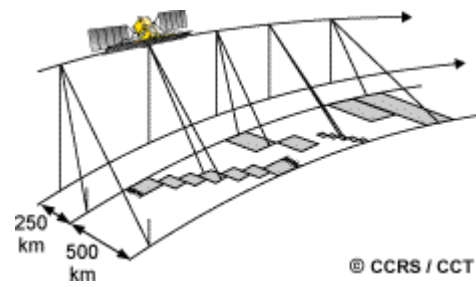
The longer L-band wavelength of JERS-1 allows some penetration of the radar energy through vegetation and other surface types.



Spaceborne SAR remote sensing took a giant leap forward with the launch of Canada's **RADARSAT** satellite on Nov. 4, 1995. The RADARSAT project, led by the Canadian Space Agency (CSA), was built on the development of remote sensing technologies and applications work carried out by the Canada Centre for Remote Sensing (CCRS)

since the 1970s. RADARSAT carries an advanced C-band (5.6 cm), HH-polarized SAR with a steerable radar beam allowing **various imaging options** over a 500 km range. Imaging swaths can be varied from 35 to 500 km in width, with resolutions from 10 to 100 metres. Viewing geometry is also flexible, with incidence angles ranging from less than 20 degrees to more than 50 degrees. Although the satellite's orbit repeat cycle is 24 days, the flexibility of the steerable radar beam gives RADARSAT the ability to image regions much more frequently and to address specific geographic requests for data acquisition. RADARSAT's orbit is optimized for frequent coverage of mid-latitude to polar regions, and is able to provide daily images of the entire Arctic region as well as view any part of Canada within three days. Even at equatorial latitudes, complete coverage can be obtained within six days using the widest swath of 500 km.

Section 3.10 Airborne and Spaceborne Radar Systems



Imaging options over a 500 km range

